Removed Duplicates Report

Every duplicate question that was removed, listed under the unique question that was kept
Grouped by your original documents

Document	Original	Removed	Unique kept	Unique questions that had duplicates
Manual Testing Interview Questions	350	116	234	63
Java Interview Questions	335	219	116	70
Java Coding Questions	304	153	151	65
Selenium Interview Questions	955	750	205	173
Selenium Coding Questions	23	2	21	2
SQL Interview Questions	97	22	75	16
API Manual Testing Questions	137	54	83	33
API Automation Questions	299	178	121	63
CI/CD Interview Questions	123	61	62	35
Git and Maven Questions	68	27	41	21
Non-Technical Interview Questions	27	3	24	3
Total	**2718**	**1585**	**1133**	**544**

How to Read This Report

Each document from your upload is one section. Inside it, questions are grouped by topic and ordered the same way as in the prioritized PDFs (Critical first). Only unique questions that had at least one duplicate are listed here.

Each entry looks like this:

[UNIQUE] Q12. What is the difference between Smoke and Sanity testing?

- Smoke vs Sanity testing?
- Difference between smoke and sanity tests? **[x2 exact repeat]**

Green [UNIQUE] line: the question that was kept in the prioritized PDF. The number (Q12) is its number in that PDF, and the tag shows its priority section and topic.

Grey lines beneath: the other wordings of the same question that were removed. A red [x2] means that exact wording appeared more than once in your document (the extra copies are counted as removed).

Note: the unique question is shown in clean, complete wording. It may be a polished version of one of the original wordings, so it can differ slightly from every line beneath it. Where an original line was only a short fragment, it is shown exactly as written in your file. Questions that had no duplicates are not shown here.

Totals: 2718 original questions, 1585 removed (177 exact repeats and 1408 same-meaning rewordings), 1133 unique kept.

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Manual Testing Interview Questions (116 removed)

Source file: Manual Testing Interview Questions. Original 350 questions, 234 unique kept, 116 removed. Prioritized PDF: Manual_Testing_Interview_Questions_Prioritized.pdf.

Defect Management & Bug Life Cycle (32 removed)

[UNIQUE] [Section 1 Critical] Q1. Explain the Defect (Bug) Life Cycle and its states.

- Bug life cycle explain
- Bug Life Cycle? **[x2 exact repeat]**
- Explain the Defect Life Cycle.
- Explain the bug life cycle?
- What is the Defect Life Cycle (Bug Life Cycle)?
- What is defect lifecycle?
- States of a defect

[UNIQUE] [Section 1 Critical] Q2. What is the difference between Severity and Priority? Give examples (high severity/low priority and low severity/high priority).

- Explain difference between severity and priority for a defect
- What is the difference between Severity and Priority?
- What is Defect Severity vs. Priority? Give examples.
- What is the difference between Priority and Severity in JIRA?
- What is a Severity vs Priority defect?
- Difference between severity and priority?
- Explain severity and priority and High severity with low priority, low severity and high priority?
- Severity vs Priority
- What is Severity and Priority?

[UNIQUE] [Section 1 Critical] Q3. How do you write a good bug report? What are the most important things to define in a bug?

- How do you write a good Bug Report?
- What is most important things to define in a bug?
- How to report a bug

[UNIQUE] [Section 1 Critical] Q4. What do you do when a developer rejects or disagrees with your defect ("Not a bug", "Rejected")? How do you convince them?

- What do you do when a developer disagrees with a bug?
- What would you do if a developer rejects your defect?
- If a bug is not accepted by developers, how do you convince them?
- How do you handle a developer who always rejects your bugs as “Not a bug”?
- What is the next step to be taken, if developer rejects the Open defect?
- How did you convince a developer it was a bug?

[UNIQUE] [Section 1 Critical] Q5. A production issue is reported that you did not catch in testing (or you missed a scenario). How do you handle it?

- If production issue is reported but you didn’t catch it in testing, how do you handle it?
- If you miss testing a scenario and it fails in production, how do you handle it?
- How do you handle production defects?
- What would you do if you find a bug in production?

[UNIQUE] [Section 1 Critical] Q7. What is the most challenging bug you have found and how did you debug it?

- What is the most challenging bug you found and how did you debug it?
- What was the most difficult bug you've found?

[UNIQUE] [Section 2 High] Q39. How do you handle priority vs severity conflicts, e.g. business says a bug is minor but you think it is critical?

- How do you handle priority vs severity conflicts in bugs?
- What if business says a bug is minor but you think it’s critical?

[UNIQUE] [Section 2 High] Q40. How do you handle a bug that is hard to reproduce or occurs intermittently? Give an example.

- Give an example of a bug that is hard to reproduce. How do you handle it?
- How do you handle intermittent issues?

[UNIQUE] [Section 2 High] Q41. How do you respond to "It's working on my machine" or "Not reproducible" from a developer or stakeholder?

- How do you handle if a developer says "It's working on my machine"?
- If a stakeholder rejects your bug saying “Not reproducible”, how will you handle?

[UNIQUE] [Section 2 High] Q42. What is Bug Triage and how is it conducted?

- What is a Bug Triage?
- Bug Triage?
- Bug triage process

[UNIQUE] [Section 2 High] Q43. What is Defect Leakage (Escape Defect)? How do you reduce it?

- What is Defect Leakage?
- What is an Escape Defect?

Smoke / Sanity / Regression & Automation Decisions (13 removed)

[UNIQUE] [Section 1 Critical] Q8. What is the difference between Smoke, Sanity and Regression testing?

- Smoke vs Sanity testing?
- What is Sanity Testing? How is it different from Smoke Testing?
- Difference between smoke and sanity tests?
- What is smoke, regression and sanity testing?
- What is the difference between Smoke and Sanity testing?
- What is the difference between a smoke and sanity suite?

[UNIQUE] [Section 1 Critical] Q9. What is the difference between Regression Testing and Retesting?

- Regression vs Retesting?
- Difference between regression and retesting?
- What is the difference between Regression and Retesting?

[UNIQUE] [Section 1 Critical] Q10. You have 250 manual test cases. On what basis will you segregate them into Smoke, Sanity and Regression suites, and when is each executed?

- There are 250 manual test cases , how will you segregate (on what basis) the regression , sanity , smoke suite?
- When Regression ,Sanity , smoke test scripts are executed ?

[UNIQUE] [Section 1 Critical] Q12. How do you decide whether a test case is a good candidate for automation? (e.g. you have 1000 test cases, which do you automate? Long-running scenario: manual or automation?)

- How will you decide that this test case is feasible or good candidate for automation ?
- If we are having 1000 test cases, what type of testing carried for automation testing?
- If there is a scenario that takes a long time to execute, would you prefer Manual Testing or Automation Testing?

[UNIQUE] [Section 1 Critical] Q13. Your regression suite takes 1.5 to 5 hours but stakeholders need results in under 1 hour. How do you reduce execution time?

- Regression suite takes ~5 hours; stakeholders need results in < 1 hour.
- If your testsuite takes 1 and half to run, what you will do to reduce the time?

[UNIQUE] [Section 1 Critical] Q14. How do you handle flaky automation tests?

- How do you handle flakiness in automation scripts?
- How do you handle flaky tests?

[UNIQUE] [Section 2 High] Q49. Your Pull Request (PR) for automation is not approved for merging. How do you handle it / convince the developer?

- What if your Pull Request (PR) is not approved for merging? How will you handle it?
- If Dev doesn't approve your PR merge for automation, how do you convince?

Test Design, Planning & Documentation (14 removed)

[UNIQUE] [Section 1 Critical] Q15. What is a Test Plan? What does it include and what are the steps to create one?

- What is a Test Plan?
- What is Test Plan?
- What is a test plan and what are the steps to create a test plan?
- What is a Test Plan and what does it include?

[UNIQUE] [Section 1 Critical] Q16. What is a Test Case and what are its key components? How is it different from a Test Scenario?

- What is a Test Case?
- What is a Test Case? What are its key components?
- What is a Test Case? How is it different from a Test Scenario?
- What is a Test Scenario?

[UNIQUE] [Section 1 Critical] Q17. How do you prioritize test cases (especially when the release deadline is short)?

- How do you prioritize test cases? **[x3 exact repeat]**
- How would you priorities Tests?
- How do you prioritize test cases when release deadline is short?

[UNIQUE] [Section 1 Critical] Q18. What are Entry and Exit Criteria in testing?

- What are Entry and Exit Criteria in Testing?
- What are Entry and Exit Criteria?
- Entry and exit criteria
- What are the inbound and outbound for testing?

[UNIQUE] [Section 1 Critical] Q21. What are Test Design Techniques?

- What are Test Design Techniques? **[x2 exact repeat]**

Situational & Behavioral Scenarios (13 removed)

[UNIQUE] [Section 1 Critical] Q25. What challenges have you faced during testing?

- What challenges do you face in Manual Testing?
- What are the challenges you have faced during testing?

[UNIQUE] [Section 1 Critical] Q26. How do you test when requirements are unclear, missing or there is no documentation?

- How do you test when requirements are unclear?
- How do you test something without requirements?
- How do you test a feature with no documentation?
- What do you do when requirements are unclear?

[UNIQUE] [Section 1 Critical] Q27. How do you manage testing under tight deadlines or release pressure?

- How do you manage testing under tight deadlines?
- How do you approach testing when timelines are very tight?
- What would you do if deadlines are too tight?
- How do you test sprint deliverables in very short time?
- What's your QA strategy in a fast-paced release cycle?
- How do you ensure quality when there's pressure to release?

[UNIQUE] [Section 1 Critical] Q28. How do you handle requirement changes during testing?

- How do you handle requirement changes during testing?
- How do you deal with changing requirements?
- How do you test when requirements keep changing?

[UNIQUE] [Section 2 High] Q60. How do you test when the test environment is unstable?

- How do you test a feature when the environment is unstable?
- What if the test environment is unstable?

[UNIQUE] [Section 2 High] Q61. Have you ever disagreed with a developer? How do you handle conflict with a developer?

- Have you ever disagreed with a developer?
- How do you handle conflict with a developer?

Agile & Scrum (12 removed)

[UNIQUE] [Section 1 Critical] Q29. What is Agile methodology?

- What is an agile methodology?
- What is Agile methodology?

[UNIQUE] [Section 1 Critical] Q30. What are the ceremonies followed in Agile/Scrum? Describe each.

- Ceremonies followed in Agile methodology?
- Describe Scrum ceremony?

[UNIQUE] [Section 2 High] Q65. What are the Scrum roles (PO, Scrum Master, Dev Team)? Who is your Scrum Master?

- What is the scrum and who is your scrum master?
- Scrum roles: PO, SM, Dev Team

[UNIQUE] [Section 2 High] Q67. What is a User Story?

- What are User Stories?
- What is a user story?

[UNIQUE] [Section 2 High] Q68. What is Acceptance Criteria? How do you test a story when the AC is unclear?

- What is Acceptance Criteria in Agile?
- Acceptance criteria
- How do you test if Acceptance Criteria (AC) is unclear in a story?

[UNIQUE] [Section 2 High] Q69. What is the Definition of Done (DoD)?

- What is a Definition of Done (DoD)?
- Definition of Done (DoD)

[UNIQUE] [Section 2 High] Q70. What is a Product Backlog? How is it different from a Sprint Backlog?

- What is a Product Backlog?
- What is backlog in Scrum methodology?
- Product backlog vs sprint backlog

[UNIQUE] [Section 2 High] Q71. What is velocity in a sprint and how is it tracked?

- Explain velocity in sprint.
- Velocity tracking

[UNIQUE] [Section 3 Medium] Q127. What is the Sprint Retrospective? What is your role in it as QA?

- What is a Sprint Retrospective?
- Retrospective meeting?
- Sprint Retrospective

Types & Levels of Testing (7 removed)

[UNIQUE] [Section 1 Critical] Q34. What is Exploratory Testing and when do you use it?

- What is Exploratory Testing? **[x2 exact repeat]**
- What is exploratory testing and when do you use it?

[UNIQUE] [Section 2 High] Q74. What is the difference between Exploratory and Ad-hoc Testing?

- What is Exploratory Testing? How is it different from Ad-hoc Testing?
- What's the difference between exploratory and ad-hoc testing?

[UNIQUE] [Section 2 High] Q75. What is Ad-hoc Testing?

- What is Ad-hoc Testing?
- What is adhoc testing?

[UNIQUE] [Section 2 High] Q78. What is White-Box Testing and how is it different from Black-Box Testing?

- What is White-Box Testing? How is it different from Black-Box Testing?
- Difference between white box and black box testing?

[UNIQUE] [Section 3 Medium] Q149. What is Usability Testing and why is it important?

- What is Usability Testing? Why is it important?
- What is usability testing in QA?

[UNIQUE] [Section 3 Medium] Q153. What is Accessibility Testing and why is it important?

- What is Accessibility Testing?
- What is accessibility testing and why is it important?

Testing Fundamentals, SDLC & STLC (7 removed)

[UNIQUE] [Section 1 Critical] Q35. What is the difference between Verification and Validation?

- Difference between Verification and Validation.
- What is the difference between Verification and Validation? **[x2 exact repeat]**
- What is difference between Validation and Verification?

[UNIQUE] [Section 1 Critical] Q36. How do you decide when to stop testing? How do you know your testing is good enough?

- How do you decide when to stop testing?
- When you decide to stop the testing?
- How do you know if your testing is good enough?

[UNIQUE] [Section 2 High] Q85. What is the Software Testing Life Cycle (STLC)? Explain its phases.

- What is the Software Testing Life Cycle (STLC)?
- What is STLC?
- Phases of STLC

Scenario-Based Feature Testing (E-commerce & Real World) (10 removed)

[UNIQUE] [Section 2 High] Q90. How do you test search functionality with filters (price, brand, category)?

- How do you test search functionality with filters (price, brand, category)?
- How do you test search functionality?

[UNIQUE] [Section 2 High] Q92. How do you test “Apply Promo Code” / coupons and discount codes?

- How do you test “Apply Promo Code” functionality?
- How do you test coupons and discount codes?

[UNIQUE] [Section 2 High] Q93. How do you test the refund flow and refund timeline in e-commerce?

- How do you test refund flow in e-commerce?
- How do you test payment refund timeline?

[UNIQUE] [Section 2 High] Q97. How do you test cross-browser compatibility?

- How do you test cross-browser compatibility?
- How do you approach cross-browser UI consistency testing?

[UNIQUE] [Section 2 High] Q99. How do you test concurrency when multiple users access the same product stock or update the cart at the same time?

- How do you test when multiple users access same product stock?
- How do you test concurrency when multiple users update cart at same time?

[UNIQUE] [Section 3 Medium] Q166. How do you test order cancellation flow?

- How do you test order cancellation flow?
- How do you test order cancellation?

[UNIQUE] [Section 3 Medium] Q168. How do you test email notifications after order confirmation?

- How do you test if email notifications are sent after order confirmation?
- How will you test email notifications for order confirmation?

[UNIQUE] [Section 3 Medium] Q176. How do you test "Add to Wishlist" functionality?

- How do you test "Add to Wishlist" functionality?
- How do you test Wishlist feature in e-commerce?

[UNIQUE] [Section 3 Medium] Q178. How do you test localization and internationalization of an e-commerce site (multi-language)?

- How will you test localization of an e-commerce site (multi-language)?
- How do you handle language/localization testing?
- How do you handle localization and internationalization testing?

Integration, DB, API, Performance & Security Scenarios (3 removed)

[UNIQUE] [Section 2 High] Q101. How do you ensure data consistency across UI, API and DB (e.g. DB data integrity after placing an order)?

- How do you ensure data consistency across UI, API, DB?
- How do you test database data integrity after placing an order?

[UNIQUE] [Section 2 High] Q102. How do you test security in login (brute force, SQL injection)?

- How do you test security in login (e.g., brute force, SQL injection)?
- How do you test SQL injection vulnerability?

[UNIQUE] [Section 2 High] Q104. How do you test without a UI (back-end or API-only features)?

- How do you test without a UI?
- How do you test features with no UI?

Metrics, Coverage & Reporting (5 removed)

[UNIQUE] [Section 2 High] Q105. What are Test Metrics?

- What are Test Metrics?
- Explain the test metrices?

[UNIQUE] [Section 2 High] **Q106. How do you report testing progress (to management / in Agile)?**

- How do you report testing progress to management?
- How do you report testing progress in Agile?

[UNIQUE] [Section 2 High] **Q107. How do you ensure maximum test coverage in less time? What is Test Coverage?**

- What is Test Coverage?
- How do you ensure test coverage?
- How do you ensure maximum coverage in less time?

[UNIQUE] [Section 3 Medium] **Q187. What is Test Case Effectiveness and how do you measure it?**

- What is Test Case Effectiveness?
- What is test effectiveness and how do you measure it?

Java Interview Questions (219 removed)

Source file: Java Interview Questions. Original 335 questions, 116 unique kept, 219 removed. Prioritized PDF: Java_Interview_Questions_Prioritized.pdf.

OOP Concepts (71 removed)

[UNIQUE] [Section 1 Critical] Q1. What is the difference between method overloading and method overriding? Give examples (e.g. from your Selenium framework).

- Overloading vs Overriding in Selenium framework?
- What is method overloading and method overriding? **[x2 exact repeat]**
- What is method overloading?
- What is method overriding?
- What is method overloading and method overriding? Provide examples. **[x2 exact repeat]**
- What is Method Overloading and Overriding?
- What is the difference between Method Overloading and Method Overriding?
- What is the method of overloading?
- Difference between overloading and overriding?
- What is method overloading vs overriding

[UNIQUE] [Section 1 Critical] Q2. What is the difference between an Abstract Class and an Interface? When do you use each?

- Abstract class vs Interface
- Difference between abstract class and interface? **[x3 exact repeat]**
- What is the difference between abstract class and interface?
- What is an interface and how is it different from an abstract class? **[x2 exact repeat]**
- Explain interfaces and abstract classes. When to use each? **[x2 exact repeat]**
- Difference between abstract and interface?
- Abstraction and Interface

[UNIQUE] [Section 1 Critical] Q3. Explain the four OOP principles (Encapsulation, Inheritance, Polymorphism, Abstraction) with real examples and where you applied them in your framework.

- OOP concepts and where you applied them in a framework
- Explain OOPs concepts with real examples.
- What are the OOP principles in Java? **[x2 exact repeat]**
- What is Object-Oriented Programming (OOP)? **[x2 exact repeat]**
- What are the main principles of OOP in Java?
- Explain OOP concepts: Encapsulation, Inheritance, Polymorphism, and Abstraction. **[x2 exact repeat]**
- What is Object-Oriented Programming (OOP) in Java?
- What is OOPS?
- What are the four main OOP principles in Java?

[UNIQUE] [Section 1 Critical] Q4. What is Polymorphism? Where do you use it?

- What is polymorphism in Java?
- Explain Polymorphism with examples.
- What is polymorphism? **[x3 exact repeat]**
- Where do you use polymorphism in java?

[UNIQUE] [Section 1 Critical] Q5. What is the difference between compile-time and runtime polymorphism (static vs dynamic binding)?

- What is the difference between compile-time and runtime polymorphism?
- What is the difference between static binding and dynamic binding?
- Polymorphism types

[UNIQUE] [Section 1 Critical] Q6. What is Encapsulation and how is it achieved? Explain with an example.

- What is encapsulation in Java and how is it achieved?
- Explain Encapsulation with an example.
- What is Encapsulation?

[UNIQUE] [Section 1 Critical] Q7. What is the difference between Abstraction and Encapsulation?

- What is the difference between abstraction and encapsulation? **[x2 exact repeat]**
- Difference Encapsulation and Abstraction ?

[UNIQUE] [Section 1 Critical] Q8. What is Abstraction and how do you implement it in Java?

- What is abstraction and how do you implement it in Java?
- Explain Abstraction with an example.

[UNIQUE] [Section 1 Critical] Q9. What is Inheritance? Explain with an example.

- What is inheritance in Java? **[x2 exact repeat]**
- Explain Inheritance with an example.
- Explain about Inheritance.
- What is Inheritance?

[UNIQUE] [Section 2 High] Q31. What is a constructor? What are default, parameterized and copy constructors, and how does it differ from a method? Does Java provide a default constructor?

- What is a constructor in Java? How is it different from a method?
- What are constructors? Explain default, parameterized, and copy constructors. **[x2 exact repeat]**
- What are Constructors in Java?
- Difference between default and parameterized constructors?
- What is the use of constructor in java?
- Will Java provide default constructor by own ?

[UNIQUE] [Section 2 High] Q32. What is constructor overloading? Can a constructor be overloaded and how does it differ from method overloading?

- Constructor overloading vs method overloading
- Can constructor be overloaded. Explain why?

[UNIQUE] [Section 2 High] Q33. What are the this and super keywords and what are they used for?

- What is the use of the `super` keyword?
- What is the `this` keyword in Java? **[x2 exact repeat]**
- Explain the “this” and “super” keywords. **[x2 exact repeat]**
- What is the ‘super’ keyword in Java?
- What is this word in Java?
- Use of this and super keywords
- What is the role of this and super keywords?

[UNIQUE] [Section 2 High] Q34. What is an abstract class, an abstract method and the abstract modifier? When would you use an abstract class?

- What is an abstract class and when would you use it?
- Abstract class vs Abstract method
- What is an abstract modifier?

[UNIQUE] [Section 2 High] Q35. How and when do you use an interface? Can a class implement multiple interfaces?

- Can a class implement multiple interfaces?
- Explain the concept of Interfaces in Java
- How and when to use interface?

[UNIQUE] [Section 2 High] **Q37. Why does Java not support multiple inheritance with classes, how is it achieved with interfaces, and what would happen if it were allowed?**

- What is the difference between single and multiple inheritance?
- Why does Java not support multiple inheritance with classes?
- How does Java achieve multiple inheritance using interfaces?
- What would happen, if multiple inheritance is possible, in Java?

[UNIQUE] [Section 3 Medium] **Q81. What is the instanceof operator and how is it used?**

- What is the “instanceof” operator and how is it used? **[x2 exact repeat]**

Collections Framework (44 removed)

[UNIQUE] [Section 1 Critical] **Q10. How does HashMap work internally? How is it implemented using key-value pairs?**

- Tell me how HashMap works internally.
- How does HashMap work internally?
- How does HashMap is implemented using key value pair?

[UNIQUE] [Section 1 Critical] **Q11. What is the difference between HashMap and HashSet?**

- HashMap vs HashSet
- Difference between HashMap and HashSet?
- Difference between Hash Map and Hash Set?

[UNIQUE] [Section 1 Critical] **Q12. What is the difference between ArrayList and LinkedList? Which is better for modification? (also: List<String> list = new ArrayList<>() vs ArrayList<String> list = new ArrayList<>())**

- What is the difference between `ArrayList` and `LinkedList`?
- Difference between ArrayList and LinkedList? **[x4 exact repeat]**
- Which is better for modification: ArrayList or LinkedList?
- Difference between Arraylist and Linked List,In which situation they are used ? Difference between List<String> list = new ArrayList<String>() and ArrayList<String> list = new ArrayList<String>();

[UNIQUE] [Section 1 Critical] **Q13. What is the difference between List, Set and Map?**

- What is the difference between `Set`, `List`, and `Map` interfaces?
- Explain the difference between List, Set, and Map. **[x2 exact repeat]**
- Difference between List and Set?
- Explain Set and Map in Java?
- What is the difference between list and set?

[UNIQUE] [Section 1 Critical] **Q14. What is the difference between HashMap, LinkedHashMap and TreeMap?**

- What is the difference between `HashMap`, `LinkedHashMap`, and `TreeMap`?
- Explain HashMap, TreeMap, and LinkedHashMap with practical examples. **[x2 exact repeat]**
- Difference between HashMap, LinkedHashMap, and TreeMap

[UNIQUE] [Section 1 Critical] **Q15. What is the difference between Comparable and Comparator?**

- What is Comparable and Comparator in Java?
- Explain Comparable and Comparator interfaces with examples. **[x2 exact repeat]**
- Comparator vs Comparable

[UNIQUE] [Section 1 Critical] **Q16. What is the Java Collections Framework? Explain the hierarchy.**

- What is a collection in Java?
- What are Collections in Java? **[x2 exact repeat]**
- Java Collections hierarchy

[UNIQUE] [Section 1 Critical] **Q17. What is the difference between an Array and an ArrayList?**

- What are the differences between array and ArrayList? **[x2 exact repeat]**
- Difference between Array and ArrayList in Java?
- Difference between Array and ArrayList?

[UNIQUE] [Section 2 High] **Q39. What is the difference between HashMap and Hashtable?**

- What is the difference between HashMap and Hashtable? **[x2 exact repeat]**
- HashMap vs Hashtable

[UNIQUE] [Section 2 High] **Q41. What is the difference between Collection and Collections?**

- Collection vs Collections difference
- Explain the Collection hierarchy and the difference between Collection and Collections. **[x2 exact repeat]**

[UNIQUE] [Section 2 High] **Q42. What is a HashMap? Can you store objects in it and how do you retrieve them?**

- What is a HashMap in Java?
- What is Hashmap? Can we store objects in Hashmap and how to retrieve them?

[UNIQUE] [Section 2 High] **Q43. Explain HashSet, TreeSet and LinkedHashSet and their differences.**

- Explain HashSet, TreeSet, and LinkedHashSet with differences. **[x2 exact repeat]**

[UNIQUE] [Section 2 High] **Q44. Explain ArrayList, LinkedList and Vector. When do you use each?**

- Explain ArrayList, LinkedList, and Vector. When to use each? **[x2 exact repeat]**

[UNIQUE] [Section 2 High] **Q45. What are Generics and how do they ensure type safety?**

- What are Generics in Java?
- What are generics in Java? How do they ensure type safety? **[x2 exact repeat]**

[UNIQUE] [Section 2 High] **Q46. How do you iterate over a Collection or Map (Iterator, for-each, Map.Entry)?**

- Explain iteration in collections.
- How to iterate over a Collection?
- Explain Map.Entry and how to iterate over Map with different methods. **[x2 exact repeat]**

[UNIQUE] [Section 2 High] **Q47. How do you remove elements while iterating (Iterator)?**

- Explain Iterator and how to remove elements while iterating. **[x2 exact repeat]**

[UNIQUE] [Section 2 High] **Q48. What is the difference between fail-fast and fail-safe iterators/collections?**

- Explain the fail-fast and fail-safe iterators. **[x2 exact repeat]**
- What are the differences between fail-fast and fail-safe collections? **[x2 exact repeat]**

[UNIQUE] [Section 3 Medium] **Q83. Explain Queue, Deque and PriorityQueue with examples.**

- Explain Queue, Deque, and PriorityQueue with examples. **[x2 exact repeat]**

Strings, Immutability & equals/hashCode (17 removed)

[UNIQUE] [Section 1 Critical] **Q18. What is the difference between String, StringBuilder and StringBuffer?**

- String vs StringBuilder vs StringBuffer
- Difference between String, StringBuffer, and StringBuilder.
- What is the difference between `String`, `StringBuilder`, and `StringBuffer`? **[x2 exact repeat]**
- Explain the difference between String, StringBuilder, and StringBuffer. **[x2 exact repeat]**
- What is the difference between String, StringBuffer, and StringBuilder in Java?
- Difference between String, StringBuilder, and StringBuffer?
- Difference between String Builder and String Buffer?

[UNIQUE] [Section 1 Critical] Q19. Why are Strings immutable in Java?

- Why strings are immutable
- What is a String in Java, and why is it immutable?
- Why String is immutable in Java?

[UNIQUE] [Section 1 Critical] Q20. What is the difference between == and .equals()?

- What is the difference between `==` and `.equals()` in Java? **[x5 exact repeat]**
- What is the difference between == and .equals()?

[UNIQUE] [Section 2 High] Q49. Why is immutability important and how do you create an immutable class?

- Why is immutability important in Java?
- Explain the concept of immutability and how to create immutable classes. **[x2 exact repeat]**

Exception Handling (19 removed)

[UNIQUE] [Section 1 Critical] Q22. What is the difference between final, finally and finalize()?

- What is the difference between final, finally, and finalize? **[x3 exact repeat]**
- Explain final, finally, and finalize() in Java.
- Difference between final, finally, finalize?
- Why do we use finally and how it differs from the final keyword?
- Difference between final, finally, and finalize()

[UNIQUE] [Section 1 Critical] Q23. What is the difference between checked and unchecked exceptions?

- What is the difference between checked and unchecked exceptions? **[x3 exact repeat]**
- Difference between Checked and Unchecked Exceptions?
- What is the difference between a checked and unchecked exceptions?
- Checked vs Unchecked exceptions

[UNIQUE] [Section 1 Critical] Q24. What is the difference between throw and throws?

- What is the use of `throw` and `throws` keywords?
- What is the difference between throw and throws in Java?
- What is the difference between throw and throws?
- Difference between throw and throws?

[UNIQUE] [Section 1 Critical] Q25. What is exception handling? Explain try-catch-finally and the types of exceptions.

- What is exception handling in Java? Explain try-catch-finally.
- What is Exception Handling in Java?
- Explain Exception Handling in Java
- Exception handling types in Java

[UNIQUE] [Section 2 High] Q50. Explain the exception hierarchy in Java and how to handle exceptions.

- Explain the Exception Hierarchy in Java and how to handle exceptions. **[x2 exact repeat]**
- Explain Exceptions hierarchy in java?

Java in Automation Frameworks (9 removed)

[UNIQUE] [Section 1 Critical] Q28. Why is Java used in automation testing and how is it integrated with Selenium / automation frameworks?

- How is Java integrated in Selenium Automation?
- How is Java used in Selenium or Automation Frameworks?
- What is Java, and why is it used in automation testing?

[UNIQUE] [Section 2 High] Q55. What are Serialization and Deserialization and how do you achieve them?

- What is Serialization in Java?
- What is Deserialization in Java?
- What is serialisation and deserialisation?
- How to achieve serialization and deserialization?

[UNIQUE] [Section 2 High] Q56. What is a POJO, why do we use it, and can you write one for serialization/deserialization?

- What do you mean by POJO why we use POJO?
- Write a POJO class for serialization/deserialization

[UNIQUE] [Section 2 High] Q61. Which design patterns do you know and how do they relate to OOP (e.g. Page Object, Singleton, Factory)?

- Design pattern in JAVA.
- What is a Design Pattern and how does it relate to OOP?

[UNIQUE] [Section 3 Medium] Q87. What is File Handling in Java? How do you read and write a file?

- What is File Handling in Java?
- How to read a file in Java?
- How to write into a file in Java?

Core Language: static, final, Access Modifiers & Basics (29 removed)

[UNIQUE] [Section 1 Critical] Q29. What is the difference between JDK, JRE and JVM?

- What is the difference between JDK, JRE, and JVM? **[x3 exact repeat]**

[UNIQUE] [Section 1 Critical] Q30. What are static variables and static methods? How do they differ from non-static members, and when should a method be static?

- What is a static variable and static method in Java?
- What are the differences between static and non-static members? **[x2 exact repeat]**
- Difference between static and non-static methods in Java.
- What are Static Methods and Variables?
- What is static in Java?
- What is static, How to set value of static variable
- What is the use of static variables?
- In which situation the method should be static and when non static?

[UNIQUE] [Section 2 High] Q64. What is the final keyword and what are its uses? (also: static vs final)

- What is the “final” keyword and its uses? **[x2 exact repeat]**
- What is final in Java?
- Use of static and final keywords

[UNIQUE] [Section 2 High] Q65. What are access modifiers in Java and their scope?

- What are access modifiers in Java?
- What are the access modifiers in Java and their scope? **[x2 exact repeat]**
- Explain Access Modifiers in Java.
- Access modifiers in java and its scope?
- What are access modifiers?

[UNIQUE] [Section 2 High] Q66. Is Java pass-by-value or pass-by-reference?

- Explain the concept of pass-by-value and pass-by-reference in Java. **[x2 exact repeat]**
- What is call by reference and call by value?

[UNIQUE] [Section 3 Medium] Q90. What are wrapper classes, autoboxing and unboxing?

- What are wrapper classes and auto-boxing/auto-unboxing? **[x2 exact repeat]**
- What is a Wrapper Class in Java?
- What is Autoboxing and Unboxing? **[x2 exact repeat]**
- Wrapper classes and autoboxing

[UNIQUE] [Section 3 Medium] Q91. What is the difference between primitive and reference data types? Is String primitive? How are data types used in automation?

- What is the difference between primitive and reference data types?
- Primitives and Non-Primitives datatypes in Java? String is primitive or non-primitive?
- What are Java Data Types, and how are they used in test automation?

[UNIQUE] [Section 4 Low] Q107. Explain conditional statements and the difference between if-else and switch-case.

- Explain Conditional Statements in Java.
- What is the difference between if-else and switch-case?

[UNIQUE] [Section 4 Low] Q108. What are loops, how are they used in QA, and what is the difference between break and continue?

- What are Loops in Java, and how are they used in QA?
- Difference between break and continue in loops?

Tricky "Can We...?" Questions (14 removed)

[UNIQUE] [Section 2 High] Q67. Can we create an object of an abstract class or instantiate an interface? Why not?

- Can we create the object for the abstract classes?
- Can we create the object for an interface?
- Can we instantiate an interface?
- Why Object creation not possible in Abstract classes?

[UNIQUE] [Section 2 High] Q68. Can we override a static or private method?

- Can you override a private or static method in Java?
- Can we override static method?
- Is it possible to override Static method?
- Can you override static or private methods?

[UNIQUE] [Section 2 High] Q69. Can the main method be overloaded or overridden? What if it is private, and can a program run without it? Why is main static?

- Can main method be overloaded?
- Can main method be overridden?
- Can we execute a java program without main method?
- If we declare the main method as private what will happen?
- Is it possible to overload main method?
- Can we over load main method in Java?
- Why is the main method static?

[UNIQUE] [Section 2 High] Q71. How do you prevent a class from being extended or a method from being overridden? (extending a final class)

- Is it possible to extend Final Class?
- How to prevent the override method in Java?

[UNIQUE] [Section 3 Medium] Q94. Can we overload static or private methods?

- Can we overload static method?
- Can we overload private methods?

Java 8: Lambda, Functional Interfaces & Streams (9 removed)

[UNIQUE] [Section 2 High] **Q73. What is a functional interface?**

- What is functional interface?
- What are functional interfaces?

[UNIQUE] [Section 2 High] **Q74. What are lambda expressions in Java 8?**

- What is a lambda expression in Java 8?
- Explain lambda expressions and functional interfaces. **[x2 exact repeat]**
- Lambda expressions in Java

[UNIQUE] [Section 2 High] **Q75. What is the Stream API? Explain filter, map, flatMap, reduce and collect.**

- What is the Stream API in Java 8?
- Explain Stream API operations: filter, map, flatMap, reduce, collect. **[x2 exact repeat]**
- What are Java Streams?

[UNIQUE] [Section 3 Medium] **Q98. Explain the Collectors class and its common methods.**

- Explain Collectors class and its common methods. **[x2 exact repeat]**

[UNIQUE] [Section 4 Low] **Q111. What is the difference between Collection.stream() and Collection.parallelStream()?**

- What is the difference between Collection.stream() and Collection.parallelStream()? **[x2 exact repeat]**

Multithreading, Synchronization & Garbage Collection (7 removed)

[UNIQUE] [Section 2 High] **Q76. What is multithreading and what is a Thread in Java?**

- What is multi-threading in Java?
- What is multithreading in Java? **[x2 exact repeat]**
- What is meant by Thread?
- Threading and Synchronization basics

[UNIQUE] [Section 2 High] **Q77. How do you handle synchronization in multithreading?**

- How do you handle synchronization in multi-threading?
- What is Synchronization in Java?

[UNIQUE] [Section 2 High] **Q78. How does Garbage Collection work? Explain memory management and memory leaks.**

- How does Garbage Collection work in Java?
- Explain memory management, garbage collection, and memory leaks in Java. **[x2 exact repeat]**

Java Coding Questions (153 removed)

Source file: Java Coding Questions. Original 304 questions, 151 unique kept, 153 removed. Prioritized PDF: Java_Coding_Questions_Prioritized.pdf.

String Programs (68 removed)

[UNIQUE] [Section 1 Critical] Q1. Reverse a string (with and without built-in methods, using StringBuilder or a loop).

- Can you write a Java program to reverse a string or check for a palindrome?
- Reverse a string using `StringBuilder`
- Write a java code for Reverse a String
- Reverse string without using reverse function
- Write a code to reverse the code, without using the built-in method.
- Java program to reverse a string
- Reverse a String (using loop and StringBuilder)
- Reverse a string **[x2 exact repeat]**
- Reverse a String without using reverse() method

[UNIQUE] [Section 1 Critical] Q2. Check whether a string is a palindrome.

- Write a java code for Check if a String is a Palindrome
- Java program to find if a string is Palindrome
- Check if a String is Palindrome
- Palindrome check for strings
- Check if a String is a Palindrome **[x2 exact repeat]**

[UNIQUE] [Section 1 Critical] Q3. Check whether two strings are anagrams.

- Write a java code for Check Anagram Strings
- Write a Java code to identify, if the pair of strings are an Anagram or not?
- Java program to determine if Two Strings are Anagrams
- Anagram program in java
- Check if two strings are anagrams **[x3 exact repeat]**

[UNIQUE] [Section 1 Critical] Q4. Count the occurrences (frequency) of each character in a string.

- WAP to find the count of each character of a string.
- Occurrences of characters in a string
- Java program to count Occurrences of Each Character in String
- Count frequency of characters
- Count occurrences of each character in a String
- Count the Occurrences of Each Character

[UNIQUE] [Section 1 Critical] Q5. Find the duplicate characters in a string.

- Duplicates in a String?
- Java program to find duplicate characters in a string
- Find duplicate characters in a String **[x2 exact repeat]**

[UNIQUE] [Section 1 Critical] Q6. Find the first (or last) non-repeating character in a string.

- Find first non-repeating character in a string (optimal solution)
- WAP to find last non repeating character from a string
- Find the first non-repeated character in a String
- Find the First Non-Repeating Character

[UNIQUE] [Section 1 Critical] Q7. Reverse the words (or each word) in a sentence, with or without split().

- Write code to reverse words in a sentence without using split()
- Write a java code for Reverse Words in a Sentence
- Java program to reverse each word of a given string
- How do you reverse the words in a sentence?
- Reverse each word in a sentence
- Reverse Words in a Sentence

[UNIQUE] [Section 1 Critical] Q8. Count vowels and consonants (and digits and spaces) in a string.

- Coding tasks: - Count vowels and consonants in a string
- Write a java code for Count Vowels in a String
- Java program to Count Vowels and Consonants in a given string
- Count vowels, consonants, digits, and spaces in a String
- Count vowels in a string
- How do you count vowels and consonants in a string?

[UNIQUE] [Section 1 Critical] Q9. Remove duplicate characters from a string (print unique characters).

- Remove duplicate letters from a string
- Java program to print unqiue characters
- Remove Duplicates from a String

[UNIQUE] [Section 1 Critical] Q10. Remove white spaces (and special characters) from a string, without built-in methods.

- Coding tasks: - Remove whitespace without built in methods
- Write a java code for Remove White Spaces from String
- Java program to remove space from a given string
- Remove all white spaces and special characters from String
- Remove All White Spaces from a String

[UNIQUE] [Section 2 High] Q28. Count the number of words in a string.

- Write a java code for Count Words in a String
- Java program to count the number of words in a string
- Find the Number of Words in a String

[UNIQUE] [Section 2 High] Q29. Count the occurrences of each word in a sentence (use HashMap) and find duplicate words.

- Count occurrences of each word in a sentence (use HashMap)
- Write a function to find out duplicate words in a given string?

[UNIQUE] [Section 2 High] Q30. Find the longest substring without repeating characters.

- Java program to find the longest without repeating characters
- How do you find longest substring without repeating characters in a string?
- Find the Longest Substring Without Repeating Characters

[UNIQUE] [Section 2 High] Q31. Separate letters, digits, uppercase and lowercase characters from a mixed string (e.g. "aBACbcEDed" to "abcde" and "ABCDE"; "Subbu123raj" to "Subburaj" and "123").

- Write code to find special character, number, capital and small letter in a given string.
- Java program to gives two Output: “abcde”, “ABCDE” for the Input String Str = “aBACbcEDed”
- Java program to gives two Output: “Subburaj”, “123” for the InputString Str = “Subbu123raj”

[UNIQUE] [Section 2 High] Q32. Compress a string (run-length encoding), e.g. "aabbcccdd" to "a2b2c3d2".

- Java program to gives Output: a2b2c3d2 for the Input String Str = “aabbcccdd”
- Compress a String (Run-Length Encoding)

[UNIQUE] [Section 2 High] **Q33. Find all permutations of a string.**

- Java program to find all permutations of a given string
- How to print all permutations of a String in Java?

[UNIQUE] [Section 2 High] **Q34. Check if one string is a rotation of another.**

- A Program to check if strings are rotations of each other or not?
- Check if a String is a Rotation of Another String

[UNIQUE] [Section 2 High] **Q35. Swap two strings without a third variable.**

- Swap string without 3rd variable?
- Java program to swap two string without using 3rd variable
- Write a java program to swap two string variables without using temp variable?

[UNIQUE] [Section 2 High] **Q36. Find the longest palindromic substring.**

- Find longest palindrome substring.
- Write a function to find out longest palindrome in a given string?
- Find the Longest Palindromic Substring

[UNIQUE] [Section 2 High] **Q37. Convert a string to an integer (atoi) and check whether a string contains only digits or is a number (regex).**

- Write a java code for Convert String to Integer
- Convert a String to an Integer (atoi Implementation)
- Check if a String Contains Only Digits
- Can you write a regular expression to check if String is a number?
- How to check whether user input is number or not in java?

[UNIQUE] [Section 2 High] **Q38. Convert lowercase to uppercase without using toUpperCase().**

- Program to convert lower to upper case without using toUppercase()?
- How Convert lower to upper case without using toUppercase() in java?

[UNIQUE] [Section 2 High] **Q39. Find the longest and the shortest word in a sentence.**

- How do you find the longest word in a sentence?
- Find the Shortest Word in a String

[UNIQUE] [Section 3 Medium] **Q97. Print even-indexed characters, or print each letter twice.**

- Java program to print even indexed characters
- Java program to print each letter twice from a given string

Number & Math Programs (27 removed)

[UNIQUE] [Section 1 Critical] **Q11. Print the Fibonacci series (up to n terms or a given range).**

- Fibonacci series program
- Write a java code for Fibonacci Series
- Write code to print the Fibonacci series?
- Java program to find Fibonacci series upto a given number range
- Write a program to print fibonacci series.
- Generate Fibonacci series up to n terms
- Print Fibonacci series

[UNIQUE] [Section 1 Critical] **Q12. Find the factorial of a number (loop and recursion).**

- Write a java code for Find Factorial of a Number
- Java program to Find Factorial on given Number
- Find factorial of a number (recursion + loop)
- Factorial using recursion

[UNIQUE] [Section 1 Critical] Q13. Check whether a number is prime.

- Write a java code for Check Prime Number
- Java program to find Prime number
- Write a program to check the given number is a prime number or not?
- Check if a number is prime
- How do you detect if a number is a prime?

[UNIQUE] [Section 1 Critical] Q14. Swap two numbers without a third variable.

- Write a java code for Swap Two Numbers Without Temporary Variable
- Java program to swap two numbers without using third variable
- Swap two numbers without third variable

[UNIQUE] [Section 1 Critical] Q15. Check whether a number is a palindrome.

- Check if a number is a palindrome
- Write code to check if a number is palindrome?
- Java program to find Palindrome number

[UNIQUE] [Section 2 High] Q42. Find the sum of digits of a number (loop and recursion).

- Write a java code for Sum of Digits
- Java program to calculate the sum of digits of a number
- How to find sum of all digits of a number in java?
- Write a program to find sum of each digit in the given number using recursion.
- Find the sum of digits of a number

[UNIQUE] [Section 2 High] Q43. Check whether a number is an Armstrong number.

- Java program to find Armstrong Number
- Armstrong number program in java
- Check if a number is an Armstrong number

[UNIQUE] [Section 2 High] Q45. Check whether a number is odd or even; count digits in a number.

- Java program to Find Odd or Even number
- Java program to find number of digits in given number

[UNIQUE] [Section 2 High] Q48. Convert a decimal number to binary, octal and hexadecimal.

- Write a program to convert Decimal To Binary, Decimal To Octal and Decimal to HexaDecimal in Java?
- Write a program to convert decimal number to binary format.

[UNIQUE] [Section 2 High] Q51. Check whether a given number is binary.

- How to check whether given number is binary or not?
- Write a program to check the given number is binary number or not?

[UNIQUE] [Section 3 Medium] Q102. Print patterns and Floyd's Triangle.

- Write a program to Printing patterns
- Write a java program to print Floyd’s Triangle?

Array Programs (35 removed)

[UNIQUE] [Section 1 Critical] Q16. Find the largest and smallest (min and max) number in an array.

- Write a java code for Find Largest Number in Array
- Largest Number in an Array?
- Write a code to get Highest number using array.
- Find the largest and smallest element in an Array **[x2 exact repeat]**
- Find Minimum and Maximum from an Array
- How to find largest and smallest number in unsorted array?
- Find the largest and smallest number in an array

[UNIQUE] [Section 1 Critical] **Q17. Find the second largest (second highest) number in an array.**

- Write a java code for Find Second Largest Number
- Find second highest number in array
- Find the second highest number in an array

[UNIQUE] [Section 1 Critical] **Q18. Remove duplicates from an array.**

- Write a java code for Remove Duplicates from Array
- Remove duplicates from an Array

[UNIQUE] [Section 1 Critical] **Q19. Find duplicate elements in an array.**

- Write a java code for Find Duplicate Elements in Array
- How to find duplicate number on Integer array in Java?
- Find duplicates in an array **[x2 exact repeat]**

[UNIQUE] [Section 1 Critical] **Q20. Find the missing number in an array (1 to n) or in a continuous sequence.**

- Write a java code for Find Missing Number in Array
- Find the missing number in an Array **[x2 exact repeat]**
- Find missing number in an integer array (1 to n)
- How do you detect missing elements in a continuous sequence?

[UNIQUE] [Section 2 High] **Q53. Find the top two maximum numbers (or the sum of the two largest) in an array.**

- Write a program to find top two maximum numbers in an array?
- How to find top two numbers from an integer array?
- Return the sum two largest integers in an array
- How to find top two numbers from an integer array in Java?

[UNIQUE] [Section 2 High] **Q54. Print the even numbers from an array; count odd and even numbers.**

- Print even numbers in an array -> `{2,4,6}`
- Write code to print only the even numbers from an array.
- Java program to count Odd and Even number from given array

[UNIQUE] [Section 2 High] **Q55. Find common elements (intersection) between two arrays.**

- Find common elements between two arrays
- Write a program to find intersection of two sorted arrays in Java?
- Compare two arrays and find common elements
- How do you find intersection elements between two arrays?

[UNIQUE] [Section 2 High] **Q56. Search for an element in an array.**

- Search element in an Array
- How to search an array to check if an element exists there?

[UNIQUE] [Section 2 High] **Q58. Find the non-repeated elements in an array, e.g. [1,1,2,2,3,4,5,5,6,6] gives [3,4].**

- Java program - input array was given [1,1,2,2,3,4,5,5,6,6], Output - [3,4] (Non-repeated elements)
- How to find first non-repeating element in array of integers?

[UNIQUE] [Section 2 High] **Q59. Find the first repeating element in an array.**

- How find the first repeating element in an array of integers?
- How do you find the first repeating element in an array?

[UNIQUE] [Section 2 High] **Q60. Move all zeroes to the end (or start) of an array, e.g. "32400121200" to "32412120000".**

- Write a java program to Move all zeroes to end of array?
- Java program to gives Output: “32412120000” for the Input String Str = “32400121200”
- Java program to gives Output: “00003241212” for the InputString Str = “32400121200”

[UNIQUE] [Section 2 High] Q61. Check whether two arrays are equal (irrespective of order).

- How do you check the equality of two arrays in java?
- How do you check if two arrays are equal (irrespective of order)?

[UNIQUE] [Section 2 High] Q64. Merge two sorted arrays.

- How to merge sorted array?
- Merge two sorted arrays

[UNIQUE] [Section 2 High] Q65. Remove a given element from an array; copy an array.

- How to remove a given element from array in Java?
- How to copy array in Java?

[UNIQUE] [Section 3 Medium] Q107. Find the sub-array with the maximum sum (maximal subarray).

- How to find sub array with maximum sum in an array of positive and negative number?
- Returns the largest sum of contiguous integers in the array
- Write a program to A maximal subarray

Collections & HashMap Programs (incl. Test Data Handling) (10 removed)

[UNIQUE] [Section 1 Critical] Q22. Remove duplicates from an ArrayList or List (using Set, keeping the order).

- Remove duplicates from an ArrayList **[x2 exact repeat]**
- How to remove duplicate elements from ArrayList in java?
- Remove duplicates from a List using Set
- Remove duplicates from a list
- How do you remove duplicate entries from a list while maintaining order?

[UNIQUE] [Section 1 Critical] Q25. Sort a HashMap by keys and by values.

- Write a program to sort a map by value?
- Sort a HashMap by values
- How do you sort a HashMap by its keys and values?

[UNIQUE] [Section 1 Critical] Q26. Sort a collection of employee objects (by ID, name, department, age) using Comparable/Comparator.

- Java Collections - Created a collection of employee records and sorted them by employee ID.
- Write a Comparator in Java to compare two employees based upon their name, departments and age?

[UNIQUE] [Section 2 High] Q66. How do you store and retrieve test data dynamically using a Map? How do you handle a key with multiple values?

- How do you store and retrieve test data dynamically using a Map?
- How do you handle key-value test data where one key has multiple values?

[UNIQUE] [Section 2 High] Q68. Sort a list alphabetically, or in reverse order.

- Sort a list alphabetically
- How do you sort out items in ArrayList in reverse direction?

Java 8 Streams Programs (3 removed)

[UNIQUE] [Section 2 High] Q74. Use Java 8 Streams to filter data (e.g. even numbers, test data) and count elements that match a condition.

- Use Java 8 Stream to filter even numbers from a list
- How do you filter test data from a list using Streams?
- How do you count elements in a stream that match a condition?

[UNIQUE] [Section 2 High] Q77. Collect stream results into a List or Set, and convert a list to a map.

- How do you collect stream results into a List or Set?
- How do you convert a list to a map using Streams?

Sorting, Searching & Complexity (4 removed)

[UNIQUE] [Section 1 Critical] Q27. Sort an array without using built-in methods (Bubble sort).

- Bubble sort algorithm
- Sort an array without using in-built method
- Sort an array without using built-in methods

[UNIQUE] [Section 3 Medium] Q118. Sort an array in place using QuickSort (recursive and iterative).

- How to sort an array in place using QuickSort algorithm?
- Write a program to Iterative Quick sort

[UNIQUE] [Section 4 Low] Q149. Implement Bucket sort and Counting sort.

- Write a program to Bucket sort
- Write a program to Counting sort

OOP, Design Pattern & Exception Coding (1 removed)

[UNIQUE] [Section 2 High] Q81. Write a Singleton class / implement the Singleton design pattern.

- Write a singleton class in java?
- Implement Singleton design pattern

File Handling Programs (1 removed)

[UNIQUE] [Section 2 High] Q88. Read a file and print its content; read a text file and count words.

- Write a java code for Read File and Print Content
- Read data from a text file and count words

Multithreading Programs (1 removed)

[UNIQUE] [Section 2 High] Q90. Write a program to create a deadlock between two threads and explain the deadlock condition.

- Write a program to create deadlock between two threads?
- Explain deadlock condition in-between two threads with example?

Linked List, Stack, Matrix & Advanced Problems (2 removed)

[UNIQUE] [Section 3 Medium] Q133. Transpose a matrix, add two matrices and multiply matrices.

- Write a program to Transpose a matrix
- Write a program to Adding two matrices in Java
- Matrix multiplication

Output & Concept Trick Questions (1 removed)

[UNIQUE] [Section 3 Medium] Q135. Can you make an array volatile? Can you pass a negative number as an array size?

- Can you make array volatile in Java?
- Can you pass the negative number as an array size?

Selenium Interview Questions (750 removed)

Source file: Selenium Interview Questions. Original 955 questions, 205 unique kept, 750 removed. Prioritized PDF: Selenium_Interview_Questions_Prioritized.pdf.

Selenium & WebDriver Fundamentals (50 removed)

[UNIQUE] [Section 1 Critical] Q1. What is Selenium WebDriver and how does it work? Explain its architecture.

- Explain WebDriver.
- Explain WebDriver Architecture (Internal Working).
- Explain the architecture of Selenium WebDriver.
- What is Selenium WebDriver and how does it work?
- What is Selenium WebDriver?

[UNIQUE] [Section 1 Critical] Q2. What is the WebDriver interface? What does WebDriver driver = new ChromeDriver() mean? Which interfaces does Selenium use and where is dynamic polymorphism seen?

- What is the WebDriver interface?
- What is Webdriver, Name methods which do not have the implementation?
- What are the methods present in the webdriver interface?
- What are the interface used in selenium?
- Consider this snippet: Web driver driver=new chromedriver(); what does the above code snippet mean?
- Where can “Dynamic Polymorphism” in Selenium WebDriver be observed?

[UNIQUE] [Section 1 Critical] Q3. What is the difference between driver.close() and driver.quit()?

- Difference between quit() and close(). **[x2 exact repeat]**
- What is the difference between driver.close() and driver.quit()?
- What is the difference between close and quit in selenium?
- Difference between driver.close() and driver.quit()
- How to close browser window?
- How to quit entire browser session?

[UNIQUE] [Section 2 High] Q66. What is Selenium and what are its core components (IDE, RC, WebDriver, Grid)?

- What is Selenium, and what are its components? **[x2 exact repeat]**
- What is Selenium and its core components?
- Difference between Selenium IDE, RC, WebDriver, and Grid

[UNIQUE] [Section 2 High] Q67. What are the limitations of Selenium?

- What are some limitations of Selenium?
- Limitations of Selenium?
- What are the limitations of Selenium?

[UNIQUE] [Section 2 High] Q68. Selenium vs Playwright: which is better and when? Can the same locators be reused?

- Selenium vs Playwright -> Which is better and when
- Is it possible to use same locators within Selenium WebDriver and PlayWright? If not then what will be your strategy in PlayWright?

[UNIQUE] [Section 2 High] Q69. What is the difference between driver.get(), driver.navigate().to() and driver.switchTo()? What are the navigation commands?

- What are the differences between driver.navigate(), driver.get(), and driver.switchTo()? **[x2 exact repeat]**
- What is the difference between driver.get() and driver.navigate().to()?
- What is difference between get() and navigate().to() in Selenium
- Different types of Navigation Commands?

[UNIQUE] [Section 2 High] **Q70. What do getText(), getAttribute(), getDomAttribute() and getDomProperty() do, and how do they differ?**

- What does getDomProperty() and getDomAttribute() methods do? What is the difference?
- Why getAttribute is used? and How its different from getDomAttribute and getDomProperty?
- What is getText() used for?
- What is getAttribute() used for?

[UNIQUE] [Section 2 High] **Q71. How do you check whether an element is displayed, enabled or selected? How do you check a disabled button without isEnabled()?**

- How to check if an element is displayed?
- How to check if an element is enabled or selected?
- How to use isDisplayed, isEnabled, isSelected?
- If the button is disabled? how to check -using getattribute()
- How to check whether web element is enabled or Disabled without using isEnabled method?
- How do you validate presence of an element?

[UNIQUE] [Section 3 Medium] **Q147. How do you open a URL, get the title, current URL and page source, navigate back/forward and refresh?**

- How to open a URL?
- How to get the page title?
- How to get current URL?
- How to get page source?
- How to navigate forward/backward?
- How to refresh page?
- How to launch a browser and open a URL in Selenium?
- What is the difference between driver.getTitle() and driver.getCurrentUrl()?
- How to verify title of page?

[UNIQUE] [Section 3 Medium] **Q148. How do you maximize, minimize and resize the browser window? What does driver.manage() return?**

- How do you maximize and resize browser windows in Selenium?
- Explain. “Driver.manage.window.maximize” (talk about option interface here)
- What is returned by driver().manage()?
- How to maximize/minimize window?

[UNIQUE] [Section 3 Medium] **Q149. What is the difference between Selenium RC and WebDriver?**

- Difference between Selenium WebDriver and Selenium RC.
- Difference between Selenium RC and WebDriver?

[UNIQUE] [Section 3 Medium] **Q150. What are the advantages of Selenium over other tools, and why do we still use it despite third-party jars?**

- What are the advantages of Selenium over other automation tools?
- Selenium uses lots of third parties jars for scripting. Then why do we still go for selenium?

[UNIQUE] [Section 3 Medium] **Q152. What is WebDriverManager?**

- What is WebDriverManager? **[x2 exact repeat]**

[UNIQUE] [Section 3 Medium] **Q153. What are DesiredCapabilities?**

- What is desired capabilities in Selenium?
- What is DesiredCapabilities?

[UNIQUE] [Section 3 Medium] **Q154. How do you verify the color, font and position of an element?**

- How to check the color of an element?
- How would you automate a scenario where you need to verify a specific color, font, and position of an element on a webpage? **[x2 exact repeat]**

Locators, XPath & CSS (55 removed)

[UNIQUE] [Section 1 Critical] Q4. What are the different types of locators in Selenium? Which do you prefer and what is their reliability order?

- Explain different types of locators in Selenium.
- Explain different locator strategies and their reliability order. **[x2 exact repeat]**
- What are Locators in Selenium?
- What are different types of locators in Selenium? Which do you prefer and why?
- Types of locators used in Selenium
- What is a selector and a locator?

[UNIQUE] [Section 1 Critical] Q5. What is the difference between findElement() and findElements()? What do they return and what happens when nothing is found?

- Difference between `findElement` vs `findElements`?
- Difference between findElement and findElements. **[x4 exact repeat]**
- Explain the difference between findElement() and findElements() methods. **[x2 exact repeat]**
- Explain the difference between findElement() and findElements().
- What is the difference between findElement() and findElements()? **[x2 exact repeat]**
- Return type of findelement and findelements?
- What error will be thrown when no element found for findelement and findelements?
- Will driver.findelements() throw an exception?

[UNIQUE] [Section 1 Critical] Q6. XPath vs CSS Selector: when do you use each, and which is faster?

- What are XPath and CSS Selector? When would you use each? **[x2 exact repeat]**
- CSS Selector vs XPath?
- Difference between XPath and CSS Selector
- Which is faster - XPath or CSS?
- Which is most faster xpath or css?
- What's the fastest locator in Selenium?
- Why is CSS locator faster than Xpath? Even though CSS is faster than Xpath ,why do 95% of the companies use XPath ?

[UNIQUE] [Section 1 Critical] Q7. What is XPath and what are its types? Difference between absolute and relative XPath.

- What is XPath, and what are its types?
- Difference between Absolute XPath and Relative XPath. **[x2 exact repeat]**
- What is the difference between Absolute and Relative XPath?
- When to use XPath? Absolute vs Relative XPath?
- What is XPath and its types?
- Difference between absolute and relative Xpath

[UNIQUE] [Section 1 Critical] Q8. How do you handle dynamic web elements (changing IDs or attributes)? Write a locator strategy.

- Handling dynamic web elements
- Write locator strategy for dynamic elements
- How do you handle dynamic web elements with changing IDs?
- How do you handle dynamic web elements?
- How would you handle dynamic web elements or elements with frequently changing IDs? **[x2 exact repeat]**
- How do you handle dynamic elements in Selenium? **[x2 exact repeat]**
- What are approached to handle dynamic WebElement?
- How to locate dynamic elements?
- How to handle dynamic elements with Xpath
- How do you handle dynamic web elements in Selenium?

[UNIQUE] [Section 2 High] Q73. Explain XPath functions and axes: contains(), starts-with(), normalize-space(), following-sibling, preceding and the :: operator.

- Use of contains(), starts-with() in Xpath
- Why would you use normalize-space() in your XPath?
- What are XPath axes like following-sibling, preceding?
- What does :: (doubles colon) in sibling xpaths represent?

[UNIQUE] [Section 2 High] Q74. What is the difference between "/" and "//" in XPath? Write XPath for an element with specific text or from given HTML.

- What is the difference between “/” and “//” in XPath?
- Write XPath for selecting element with specific text
- Write XPath expressions from given locations
- In selenium, if you want to access the element that has the text “This element has an ID that changes every time the page is loaded” in it, then which of the following will you use?

[UNIQUE] [Section 2 High] Q75. How do you select the nth (e.g. third) element among multiple similar elements using XPath and CSS?

- How to get n-th element using XPath and CSS?
- In an HTML with multiple similar elements, how do you select the third one?
- Handling multiple elements with similar attributes
- If their are multiple similar elements then How does WebDriver Identifies them?

[UNIQUE] [Section 2 High] Q76. How do you handle shadow DOM elements (and iframe switching)?

- How to handle shadow DOM & iframe switching
- How you will handle elements which are inside shadow dom
- How to interact with shadowRoot? (use the updated version of webdriver to answer)
- How do you handle shadow DOM elements in Selenium?

[UNIQUE] [Section 3 Medium] Q156. If proper XPath, CSS and ID are not available, how do you identify an element? What are the CSS selector attributes?

- If proper Xpath, CssSelector and ID are not available, how do you identify an object?
- Attributes of CSS Selector?

Waits & Synchronization (40 removed)

[UNIQUE] [Section 1 Critical] Q9. What are the different waits in Selenium? Explain implicit vs explicit vs fluent wait and when to use each.

- Different wait strategies and when to use them
- Difference between implicit, explicit, fluent waits
- What are the different waits in Selenium - and when do you use implicit, explicit, or fluent waits?
- Implicit vs Explicit wait?
- What are different types of waits in Selenium? **[x4 exact repeat]**
- What is implicit wait vs. explicit wait vs. fluent wait in practical scenarios? **[x2 exact repeat]**
- What are Waits in Selenium, and what types are there?
- Difference between Implicit Wait and Explicit Wait.
- Explain different types of Waits in Selenium.
- Difference between Implicit, Explicit and Fluent waits in Selenium?
- Types of waits in Selenium?
- What is the difference between implicit, explicit, and fluent wait?
- What are different types of waits in Selenium WebDriver?

[UNIQUE] [Section 1 Critical] Q10. Can implicit and explicit waits be mixed? Which one takes effect and is it good practice?

- If both wait method that is implicit and explicit is mentioned in the script, then which one is work? is it good practice to mention both in good?
- Can waits be mixed?

[UNIQUE] [Section 1 Critical] Q11. How do you handle AJAX, slow-loading pages or elements that appear late (dynamic waits, page load)?

- How do you handle dynamic waits in a slow-loading web application?
- How do you handle AJAX calls in Selenium?
- How would you handle a scenario where an element takes variable time to load?
- Pages use heavy AJAX; elements appear late and flake.
- How to handle AJAX elements?
- Wait Until Page Load Completes
- How to wait for elements dynamically?
- How do you handle dynamic timeouts in test execution?
- How do you handle waits and synchronization in your framework?

[UNIQUE] [Section 2 High] Q78. What is synchronization in Selenium?

- Explain about synchronization in selenium?
- What is synchronization?
- Explain synchronization in Selenium

[UNIQUE] [Section 2 High] Q79. Pros and cons of implicit and explicit waits. Why prefer explicit over fluent? What is the implicit wait best practice?

- Pros and cons of Implicit wait and Explicit wait.
- Why we prefer explicit wait instead of fluent wait? What are the disadvantages of fluent wait?
- Which concept they have implemented in explicit and fluent wait?
- What is implicit wait timeout best practice?

[UNIQUE] [Section 2 High] Q80. What is the default polling time of explicit/fluent wait?

- What is default polling time in explicit wait and in implicit wait?
- Default polling time in FluentWait?

[UNIQUE] [Section 2 High] Q81. What are Expected Conditions? Give 5 examples. How do urlToBe() and titleIs() differ?

- What are Expected Conditions in Selenium WebDriverWait and give 5 examples? **[x2 exact repeat]**
- How does WebDriverWait for urlToBe() and TitleIs() differs?

[UNIQUE] [Section 2 High] Q82. Write code for implicit, explicit and fluent waits, and a custom FluentWait instead of Thread.sleep. Thread.sleep vs implicit wait?

- Implicit Wait example:
- Explicit Wait example:
- Fluent Wait? Give an example
- FluentWait Instead of Thread.sleep
- Write custom wait using FluentWait
- Thread.sleep vs Implicit Wait?
- When to use Explicit Wait?

[UNIQUE] [Section 3 Medium] Q159. How do you define page load strategies? What is Page Load Timeout?

- How to define page load strategies?
- What is Page Load Timeout?

Element Interactions, Actions & JavaScriptExecutor (84 removed)

[UNIQUE] [Section 1 Critical] Q12. How do you handle dropdowns (Select class: visible text, index, value; get all options; deselect)?

- How do you handle dynamic dropdowns and select options using Selenium WebDriver?
- How do you handle dropdowns in Selenium? **[x3 exact repeat]**
- How to handle dropdowns?
- Select dropdown by visible text:
- Select dropdown by index:
- Select dropdown by value:
- How to handle dropdowns in Selenium
- How to handle dropdown menus in Selenium?
- How to deselect a selected option?
- How to get values from dropdowns?
- How to get all dropdown options?

[UNIQUE] [Section 1 Critical] Q13. What is the Actions class and when do you use it? Actions vs direct WebElement methods.

- What is the difference between Actions class and direct element methods? **[x2 exact repeat]**
- What is the Actions class in Selenium, and when do you use it?
- How do you perform keyboard and mouse interactions using Selenium? **[x2 exact repeat]**
- What is the difference between click() function in webelement interface and click() function in Actions class?

[UNIQUE] [Section 1 Critical] Q14. What is JavaScriptExecutor and why/how is it used?

- How do you execute JavaScript in Selenium? **[x2 exact repeat]**
- What is JavaScriptExecutor in Selenium, and why is it used?
- What is JavaScriptExecutor? **[x2 exact repeat]**
- What is JavaScriptExecutor and how is it used?

[UNIQUE] [Section 1 Critical] Q15. How do you scroll a page, scroll to an element that is not in view, or scroll until the footer is visible?

- Scenario: You need to scroll to an element that’s not in view. How do you do it?
- How would you automate a scenario where you need to verify if a user is able to scroll down a webpage until the footer section is visible? **[x2 exact repeat]**
- How do you scroll a page in Selenium?
- How do you scroll in Selenium?
- How to scroll down a page?
- Scroll Until Element Visible
- Scroll page using JSExecutor:
- Scroll into view using JavaScriptExecutor

[UNIQUE] [Section 1 Critical] Q16. How do you capture a screenshot on test failure (named by test) and attach it to the report?

- Capture screenshot on failure
- How do you capture screenshots on test failures and attach them to Extent Reports?
- How do you capture screenshots on failure?
- How do you take screenshots only on test failure?
- How can you take the screenshot for the failed testcases?
- Screenshot with Failed Test Name
- Capture and save screenshots on failure
- How do you capture logs and take screenshots when tests fail?
- How would you take screenshots and generate visual reports in Selenium? **[x2 exact repeat]**

[UNIQUE] [Section 2 High] Q83. How do you handle dynamic dropdowns (values change by previous selection, 500 values, last option, Streams)?

- Scenario: Dropdown values change based on previous selection. How to handle it?
- Dynamic dropdown with 500 values - will you loop?
- Dynamic Dropdown Using Streams
- Click last option in the dropdown (Last drop-down changes dynamically)

[UNIQUE] [Section 2 High] Q84. How do you handle auto-suggest / auto-fill fields and validate the listed options?

- How do you manage AutoFill Fields and validate the options listed?
- Handle auto-suggest dropdowns

[UNIQUE] [Section 2 High] Q85. How do you perform mouse hover, double-click and right-click?

- How to perform mouse hover?
- How to double-click?
- How to right-click?
- How to perform mouse hover using Actions class
- How to click right click of mouse?
- How to perform mouse hover or drag-and-drop?

[UNIQUE] [Section 2 High] Q86. How do you automate drag and drop?

- Scenario: How do you automate drag and drop?
- How do you perform Drag and Drop in Selenium?
- How to drag and drop?

[UNIQUE] [Section 2 High] Q87. How do you handle keyboard events (ENTER key, keyboard shortcuts)?

- How to handle keyboard events?
- How to simulate ENTER key?
- Scenario: How do you test keyboard shortcuts in a web app?

[UNIQUE] [Section 2 High] Q88. Why do we use build() and perform() with Actions? Can perform() be used without build()?

- Why do we have to use build() and perform() with the action object?
- Can we use perform() along in scripting without build()
- What is difference between build() and perform() in selenium?

[UNIQUE] [Section 2 High] Q89. How do you handle hidden or invisible elements?

- Scroll actions and handling hidden elements
- How do you handle hidden elements in Selenium? **[x2 exact repeat]**
- Different ways to handle hidden elements?
- How to handle invisible elements?

[UNIQUE] [Section 2 High] Q90. How do you handle web tables (dynamic data, extract data, verify sorting)?

- How would you automate a table with dynamic data?
- How to handle web tables whose values change dynamically?
- Verify Sorting in Table
- Extract table data dynamically

[UNIQUE] [Section 2 High] Q91. How do you handle date pickers / calendars?

- How do you handle date pickers in Selenium?
- How to handle calendar date pickers?

[UNIQUE] [Section 2 High] **Q92. How do you handle SSL certificate errors?**

- How do you handle SSL certificate errors using Selenium WebDriver?
- Explain how to handle SSL certificate errors in Selenium. **[x2 exact repeat]**
- How do you handle SSL certificate errors in Selenium?
- How do you handle SSL certificate errors?
- How to handle SSL certificates?

[UNIQUE] [Section 2 High] **Q93. How do you take a screenshot? Is TakesScreenshot an interface or a class?**

- How do you capture screenshots in Selenium? **[x2 exact repeat]**
- How do you take screenshots in Selenium?
- How to capture screenshot?
- How to take screenshots in Selenium?
- Does takes screenshot is interface or class

[UNIQUE] [Section 3 Medium] **Q160. When should you use sendKeys() vs sendKeys() from the Actions class?**

- When should you use sendKeys() and sendKeys() from Actions Class?
- Why do we have 2 different sendKeys? One for action and another in Actions?

[UNIQUE] [Section 3 Medium] **Q161. How do you handle checkboxes and radio buttons (e.g. click all checkboxes)?**

- How do you handle checkboxes and radio buttons?
- Click all checkboxes on a page

[UNIQUE] [Section 3 Medium] **Q162. How do you validate infinite scroll and lazy-loaded content?**

- Scenario: Page has infinite scroll. How to validate all items load?
- Scroll and validate lazy-loaded content

[UNIQUE] [Section 3 Medium] **Q163. How do you handle cookies?**

- How do you handle cookies in Selenium?
- How to handle cookies?

[UNIQUE] [Section 3 Medium] **Q164. How do you verify tooltips?**

- How do you verify tooltips in Selenium?
- How do you verify tooltip text?

[UNIQUE] [Section 3 Medium] **Q165. How do you capture network traffic, simulate network conditions and read browser console logs?**

- How to simulate network conditions in Selenium WebDriver?
- How do you capture network traffic in Selenium?
- How do you capture logs from the browser console?
- Capture Browser Console Errors

[UNIQUE] [Section 3 Medium] **Q167. How do you get dropdown values and print them in ascending order (TreeSet)?**

- Get the values from the dropdown and print them in Ascending order
- Using TreeSet to find elements command

[UNIQUE] [Section 3 Medium] **Q168. How do you take element-only and full-page screenshots?**

- How to take screenshot of element only?
- How do you take full-page screenshots in Selenium?

[UNIQUE] [Section 4 Low] **Q196. How do you highlight an element before clicking (JavaScript)?**

- Highlight Element Before Click
- Highlight element using JS:
- Highlight a WebElement

Windows, Frames, Alerts & Pop-ups (47 removed)

[UNIQUE] [Section 1 Critical] Q17. How do you handle alerts, pop-ups and browser dialogs (accept, dismiss, getText, sendKeys)?

- Handling alerts and multiple tabs
- How do you handle alerts, popups, and iframes?
- Scenario: How do you handle alerts?
- How would you handle pop-up windows or alert boxes in your Automation script? **[x2 exact repeat]**
- How do you handle pop-ups, alerts, and browser dialogs? **[x2 exact repeat]**
- How do you handle popups in Selenium?
- Explain how you handle Alerts, Pop-ups, and Multiple Windows.
- Handle JS alerts, confirm dialogs, and in-DOM modals.
- How do you handle Alert in Selenium?
- How to handle Alerts in Selenium?
- Describe how to handle the below items using selenium -iframe -windows -table -Alerts
- How to handle alerts?
- Accept or dismiss alert?
- Handling alerts and JavaScript popups
- How to handle alerts and pop-ups?
- How do you handle alerts and pop-ups in Selenium WebDriver?

[UNIQUE] [Section 1 Critical] Q18. How do you handle multiple windows or tabs and switch between them (parent window, title check)?

- What happens when there are multiple browser windows open, How do you switch between them?
- How to handle multiple windows/tabs?
- How do you handle multiple browser windows in Selenium?
- Window Handling in Selenium -Switching from another window to Parent window
- Window Handling (Check Title, Then Switch)
- How to switch between windows?
- How to switch to popup window?
- How to handle multiple tabs?
- Switch between multiple tabs
- How to handle multiple browser windows or tabs?
- How do you handle multiple browser windows or tabs?
- How to get window handle?

[UNIQUE] [Section 1 Critical] Q19. How do you handle frames and iframes (switch in, switch back, nested frames)?

- Handling iframes
- Explain how you can switch back from a frame.
- What is the difference between switchTo() methods and how do you switch between frames and windows? **[x2 exact repeat]**
- How do you handle frames and iframes in Selenium?
- How do you handle elements inside multiple nested iframes?
- How to handle frame in Selenium?
- How to handle frames?
- How to return to main content?
- How to switch between frames and iframes
- How to handle frames and iframes?

[UNIQUE] [Section 2 High] Q94. What are the return types of getWindowHandle() and getWindowHandles()?

- What is the return type of getWindowHandles()?
- Return type of getwindowhandle() and getwindowhandles()?
- What will driver.getWindowHandels() return?

[UNIQUE] [Section 2 High] Q95. Pop-ups appear at unpredictable times. How do you handle them with WebDriver?

- In a web page, there are several Pop-up, but we don't when the pop-up will appear, in this case how we will handle the Pop-up using Selenium WebDriver (Java)
- On a web page, there are several Pop-up, but we don't when the pop-up will appear, in this case how we will handle the Pop- up using Selenium WebDriver (Java)

[UNIQUE] [Section 2 High] Q97. How do you handle Windows-based pop-ups and Windows applications (AutoIt, Robot)?

- How do you handle Windows-based pop-ups?
- What if Windows popup occurs during test execution and due to that can't execute automated tests, how u will resolve this error?
- How will you automate Windows based application?
- Can Selenium handle Windows popups?

[UNIQUE] [Section 3 Medium] Q170. How do you open a new blank window/tab, close only the current tab or close all tabs?

- How to move from one window to another and create a new blank window (tab)?
- How to close only current tab?
- How to close all tabs?

[UNIQUE] [Section 3 Medium] Q171. What is the difference between frames and iframes and how are they identified?

- How does frames and iFrames are identified?
- Difference between frames and iframes?

Exceptions & Troubleshooting (20 removed)

[UNIQUE] [Section 1 Critical] Q20. What is StaleElementReferenceException? What causes it and how do you handle or avoid it (after refresh or AJAX)?

- How do you handle StaleElementReferenceException? **[x2 exact repeat]**
- How do you handle stale element exceptions in Selenium?
- How do you handle a stale element after a page refresh?
- Frequent StaleElementReferenceException after AJAX updates.
- What is StaleElementReferenceException and how do you handle it? **[x2 exact repeat]**
- Types of Exceptions and how to handle stale element exception?
- How do you Handling StaleElementReferenceException
- What is a stale element reference exception?
- How to avoid stale element exception?
- How to handle StaleElementReferenceException?

[UNIQUE] [Section 1 Critical] Q21. What are the common exceptions you faced in Selenium and Java, and which was hardest to solve?

- List down the top exceptions and how you solved them in Selenium WebDriver?
- Which Selenium exception was difficult to handle?
- State some exception which you have faced in your framework? (Don't mention only selenium explain. Explain java exception also)
- Common exceptions in Selenium?

[UNIQUE] [Section 1 Critical] Q22. How do you avoid NoSuchElementException (with and without try/catch)? Intermittent NoSuchElement issues?

- How can you bypass the issue of Element Not Found without using Try catch or throws or custom exceptions?
- How to avoid NoSuchElementException without using try/catch block and with try/catch block?
- Intermittent NoSuchElement/ElementNotInteractable issues.

[UNIQUE] [Section 1 Critical] Q23. The locator is correct and the element is on the page, but you still get "element not found". How do you resolve it?

- Error is throwing as Element not found but when I go and check that element is available in the web page? The element is not hidden so no need to use Java script executor? How do you solve this?
- If we are using correct locator but still getting element not found error then how you will resolve this error?

[UNIQUE] [Section 1 Critical] Q24. What causes ElementNotInteractable / ElementClickIntercepted exceptions, and what do you do when click() does not work?

- Scenario: You get 'element not interactable' error. What do you do?
- What would you do if click() is not working on a button?
- How do you handle “ElementClickInterceptedException” in Selenium?
- What causes ElementNotInteractableException?

Flaky Tests, Retry & Debugging (32 removed)

[UNIQUE] [Section 1 Critical] Q25. How do you handle and avoid flaky tests? What are the common causes and how do you investigate them (CI, Jenkins, BrowserStack)?

- Strategies to reduce flaky tests
- Dive Deep -> Debugged flakiness root cause with HAR logs
- What’s your approach to debugging flaky or intermittent failures?
- In one of your Selenium?WebDriver suites you noticed many flaky tests (sometimes pass, sometimes fail) especially on the UI side when executed in a CI pipeline (e.g., on Jenkins + Browser Stack). Describe how you would investigate and resolve flakiness.
- Scenario: How do you deal with flakiness in tests?
- How do you avoid flaky tests in Selenium?
- How do you ensure your automation scripts are stable and not flaky?
- How do you handle flaky tests in your framework? **[x2 exact repeat]**
- How do you handle flaky tests and synchronization failures?
- What are common causes of flaky tests?
- What is flaky test analysis and how do you conduct it?
- How do you deal with unstable test scripts in CI?

[UNIQUE] [Section 1 Critical] Q26. How do you implement retry logic for failed tests (IRetryAnalyzer, retry a step in the same test)?

- RetryAnalyzer implementation
- How to run flaky test cases? (with code)
- Scenario: How do you retry a failed step in the same test?
- Explain retry analyzer?
- Retry Failed Test Cases (TestNG)
- Retry failed test case
- How do you implement retry logic in test cases?
- Retry failed tests

[UNIQUE] [Section 1 Critical] Q27. If a test fails, what is your next step? A script that worked yesterday fails today. How do you debug?

- If a Test fails, what will be your next step? **[x2 exact repeat]**
- A script you wrote was working fine yesterday but is failing today. How do you troubleshoot? **[x2 exact repeat]**
- What’s your approach to debugging a failed Selenium test?
- How to debug failed tests?

[UNIQUE] [Section 1 Critical] Q28. Tests pass locally but fail on Jenkins/CI nodes. How do you debug?

- Tests pass locally but fail on Jenkins (CI) nodes.
- Your tests pass locally but fail in CI/CD. How do you debug?

[UNIQUE] [Section 1 Critical] Q29. The application has minor changes (small CSS/HTML changes break many tests). How do you maintain and update scripts?

- If the application has minor changes, what would be your approach to modifying the Automation scripts? **[x2 exact repeat]**
- Your Automation script is failing due to a change in the application. How do you handle this? **[x2 exact repeat]**
- Small CSS/HTML changes cause many test failures.
- How do you maintain test scripts for a changing application?

[UNIQUE] [Section 1 Critical] Q30. Your automation scripts are slow, or the regression suite takes about 5 hours while stakeholders need results in under 1 hour. How do you speed it up?

- Your Automation scripts are running slowly. How can you improve the speed ? **[x2 exact repeat]**
- Regression suite takes ~5 hours; stakeholders need results in < 1 hour.

File Upload/Download & Broken Links (26 removed)

[UNIQUE] [Section 1 Critical] Q31. How do you handle file upload in Selenium (sendKeys, Robot class, AutoIt)? Upload and download together?

- Can you automate file upload/download in Selenium? What tools or workarounds do you use?
- How do you handle file uploads and downloads in Selenium? **[x2 exact repeat]**
- How do you handle file uploads in Selenium?
- How do you test file uploads and downloads in Selenium?
- File Upload (AutoIt or Robot)
- How to upload a file?
- Handle file upload with Selenium

[UNIQUE] [Section 1 Critical] Q32. How do you handle file download and verify it (file exists, content is correct)?

- Scenario: You need to check if a file is downloaded. How?
- How would you automate a scenario where you need to validate the contents of a downloaded file after clicking a button on a webpage? **[x2 exact repeat]**
- How do you handle file downloads in Selenium?
- Validate file download end-to-end.
- How to download a file using Selenium?
- File Download Verification
- How do you handle file downloads using Selenium?

[UNIQUE] [Section 1 Critical] Q33. How do you find broken links and broken images on a page?

- How to fetch all the broken links with its name and url?
- Fetch all links from a website & check broken ones
- How do you validate broken links in Selenium?
- How do you handle broken images on a web page?
- How do you verify broken links on a web page?
- How would you check the broken links, in the webpage?
- Ways to find broken links in Selenium?
- Capture Broken Links
- How to check broken links?
- Capture all links and validate them
- Validate broken links

[UNIQUE] [Section 2 High] Q101. How do you handle file upload when the element is not type=file, or in remote execution (Grid/cloud)?

- How to handle file upload when type attribute does not file for upload web element.
- How do you test file uploads in remote execution (like Grid or Cloud)?

[UNIQUE] [Section 3 Medium] Q175. How do you count the links on a page (by tag) and handle hidden links?

- How to calculate links on a page? (Answer with HTML tag)
- When should you find a URL and validate its existence on a web page? What are hidden links?

Page Object Model & Page Factory (34 removed)

[UNIQUE] [Section 1 Critical] Q34. What is the Page Object Model (POM)? What are its advantages and how do you implement it (e.g. a LoginPage class)?

- Explain the Page Object Model (POM) - can you write a sample LoginPage.java using best practices?
- Explain Page Object Model (POM).
- Explain the Page Object Model (POM) pattern and its advantages. **[x2 exact repeat]**
- What is the Page Object Model (POM) and why is it important? **[x2 exact repeat]**
- What is Page Object Model (POM) in Selenium?
- How do you create a Page Object Model (POM) in Selenium?
- How do you implement Page Object Model in your framework?
- What is the Page Object Model (POM)?
- What are the benefits of POM?
- What is Page Object Model (POM)? **[x3 exact repeat]**
- Advantages of POM?
- What is the Page Object Model (POM) in Selenium?
- What are the main advantages of using POM?
- How do you implement the Page Object Model in Selenium?
- How does POM support test maintenance?
- How do you organize a framework using POM structure?

[UNIQUE] [Section 1 Critical] Q35. What is the difference between Page Object Model and Page Factory?

- Page Object Model vs Page Factory (real-world differences)
- Difference between page object vs Page factory
- What is the difference between POM and Page Factory?

[UNIQUE] [Section 1 Critical] Q36. What is Page Factory? @FindBy vs driver.findElement, initElements and the annotations used in POM.

- @FindBy vs driver.findElement
- PageFactory usage
- What is PageFactory in Selenium?
- What is the Page Factory pattern?
- What is page object and page factory model?
- What is Page Factory in POM Design Pattern?
- Annotation used in Page Object Model?
- What is PageFactory?
- What is the Page Factory in Selenium?

[UNIQUE] [Section 2 High] Q102. How do you initialize web elements in POM? What exception occurs if you do not? What do you write first in a page class?

- How do you initialize web elements in POM? What error or exception will come if not initiated?
- On page Object Model Framework (POM), how do you initialize the elements of a page to be used in the runner class? (name of the PageObjects class is “”SignupPage.java”” and the dirver object name is “”driver””).
- In Page object model once you create loginpage.java class what is the first thing you start with writing initially. How are you initiating writing something into a page class?

[UNIQUE] [Section 2 High] Q103. You have 10 pages. How do you achieve POM, call page methods from tests and use it with TestNG or Cucumber?

- Suppose you have 10 pages in your application then how to achieve POM. What you will do?
- How do you call methods from the Page Class in the Test Class?
- Can you use POM with other frameworks like TestNG or Cucumber?
- How do you handle multiple pages using POM?

Framework Design & Architecture (41 removed)

[UNIQUE] [Section 1 Critical] Q37. Explain your automation framework (architecture, folder structure and flow).

- Explain your automation framework structure.
- Explain the structure of an automation framework
- Explain your automation framework.
- Explain your current automation framework architecture.
- Which Automation framework you are using in your organization can you tell me the folder structure and flow of your framework?
- Explain the concept of test automation framework architecture.
- How should you structure your automation framework folders and files?
- How do you organize your test framework folder structure?
- Folder structure in a Selenium framework

[UNIQUE] [Section 1 Critical] Q38. How would you design a scalable, maintainable framework (Java + TestNG/Cucumber, UI and API, Docker, CI/CD) or build one from scratch? Best practices?

- Problem: Design a scalable automation framework for a travel booking platform.
- How would you design a scalable Selenium automation framework using Java + TestNG?
- What strategies do you use to make your test scripts reusable and maintainable as the project scales?
- In your automation framework you need to support both UI and API tests, across browsers and mobile devices, and integrate with Docker containers and CI/CD pipelines. Describe how you would design or enhance your framework (using Java + TestNG/Cucumber) to support this.
- What are best practices for creating scalable and maintainable test automation frameworks?
- What strategies you followed while building a selenium framework from scratch?
- How do you make your framework scalable?
- How to make your framework scalable
- What are the best practices for organizing test code?

[UNIQUE] [Section 1 Critical] Q39. What is an automation framework, why do you need one, and what types exist (data-driven, keyword, hybrid, BDD)?

- What is an automation framework, and why is it important in Selenium projects?
- What are the types of Selenium automation frameworks?
- What types of automation frameworks have you worked with?
- What is a test automation framework, and why do you need one?
- What are the main types of test automation frameworks?
- What is a test automation framework? **[x2 exact repeat]**
- Why is it important to use a framework?
- What are some popular automation frameworks?
- Why is framework design important for automation?
- Types of frameworks: POM, hybrid, data-driven
- Common frameworks used with Selenium?

[UNIQUE] [Section 1 Critical] Q40. What was the most challenging thing in designing your framework, and what automation challenges did you overcome?

- What is most challenging thing you faced while designing framework ?
- What are the challenges you faced in automation and how did you overcome them?
- What are the challenges you have faced during testing?

[UNIQUE] [Section 2 High] Q104. What is a hybrid framework and how do you create one?

- How do you create a Hybrid Framework in Selenium?
- What is a hybrid framework?

[UNIQUE] [Section 2 High] Q107. Which design patterns have you used (Singleton, Factory, POM)? Where do you use Singleton? Factory vs Singleton?

- Explain Singleton design pattern in automation
- Where do you perform the singleton design pattern? If you don't use it, then do you have an idea about this?
- Difference between Factory design and Singleton framework?
- What are design patterns and which ones apply to test automation?

[UNIQUE] [Section 2 High] Q108. How do you build reusable components, utility classes and modular tests?

- What are reusable components in a framework?
- What is modularity in a test framework?
- Reusable utility classes
- Modularizing your tests

[UNIQUE] [Section 2 High] Q109. How do you integrate API and database testing with UI automation? API vs UI frameworks?

- How do you integrate API testing with Selenium UI tests?
- How do you perform database testing in your automation framework?
- What is the difference between API testing and UI testing frameworks?

[UNIQUE] [Section 3 Medium] Q176. What are the SOLID principles and how do you apply them in automation?

- What are SOLID principles and how do you apply them in automation?
- What are the SOLID principles and how do they apply to test automation?

[UNIQUE] [Section 3 Medium] Q177. What is the difference between a framework and a library?

- What is the difference between a framework and a library?
- What is the difference between a library and a framework?

[UNIQUE] [Section 3 Medium] Q178. How do you implement error handling and recovery in the framework?

- How do you implement error handling and recovery in your framework?
- Exception handling in frameworks

TestNG (112 removed)

[UNIQUE] [Section 1 Critical] Q41. What is TestNG and why use it? How does it differ from JUnit?

- What is TestNG and how does it differ from JUnit? **[x2 exact repeat]**
- What is TestNG, and why is it used with Selenium?
- What is TestNG, and why is it preferred over JUnit for Selenium?
- What is the difference between JUnit and TestNG in Selenium?
- What is TestNG and what are its advantages over JUnit?
- What is the importance of the testng framework?
- Why we use TestNG in your framework
- What is TestNG?
- What is TestNG and why use it?
- TestNG vs Junit
- What is TestNG and why is it used in automation testing?

[UNIQUE] [Section 1 Critical] Q42. Explain TestNG annotations and their execution order (BeforeSuite, BeforeTest, BeforeClass, BeforeMethod). BeforeTest vs BeforeMethod.

- Difference between @BeforeTest and @BeforeMethod.
- Explain TestNG annotations: @BeforeClass, @BeforeMethod, @AfterClass, @AfterMethod. **[x2 exact repeat]**
- Explain @BeforeSuite, @AfterSuite, @BeforeTest, @AfterTest annotations **[x2 exact repeat]**
- Explain execution order: BeforeTest, BeforeMethod, AfterMethod, AfterTest.
- Explain the TestNG annotations used in Selenium.
- Explain TestNG annotations and their execution order.
- What are different testng annotations?
- Explain the difference between beforemethod, beforetest, and beforeclass
- List out the testng annotation hierarchy order?
- Difference between after suite and before suite?
- Annotations used in TestNG
- What are the main annotations used in TestNG?
- What is the difference between @BeforeTest and @BeforeMethod?

[UNIQUE] [Section 1 Critical] Q43. How do priorities work in TestNG? How do you control execution order?

- How do priorities work in TestNG?
- What is the role of @Test(priority) in TestNG?
- Test Order in TestNG?
- How to prioritize test cases in TestNG?
- Execution order control
- Prioritization of test cases
- How do you prioritize tests in TestNG?

[UNIQUE] [Section 1 Critical] Q44. What is DataProvider and how do you use it for parameterization (and parameters through XML)? Attributes of @DataProvider.

- What is DataProvider in TestNG and how to use it for parameterization? **[x2 exact repeat]**
- What are the attributes of @DataProvider annotation? **[x2 exact repeat]**
- How you used @DataProvider in TestNG? How did you apply parameterization?
- What is @dataprovider? **[x2 exact repeat]**
- Use of DataProvider
- What is DataProvider in TestNG?
- Explain DataProvider and Scenario Outline.
- What is parameterization?
- Parameterization using XML
- How do you perform parameterization in TestNG?

[UNIQUE] [Section 1 Critical] Q45. What is testng.xml used for? How do you configure and run tests, add or remove tests, and run multiple suites?

- What is TestNG XML configuration file and how to run tests? **[x2 exact repeat]**
- What is TestNG XML file used for?
- What is the purpose of testing XML
- Some questions in TestNG XML?
- How can you configure tests in testng?
- How to add/remove test cases in Testng.xml?
- What is the use of testng.xml?
- How many suits can be there in testNG , what if I run all the suits?
- In testNG, do we have multiple suite in one XML file and what If I want to run all suits?
- TestNG suite XML configuration

[UNIQUE] [Section 1 Critical] Q46. How do you run tests in parallel in TestNG (parallel attribute, testng.xml or Maven) and make the framework thread-safe?

- Parallel execution using testng.xml
- What's your approach to running parallel tests in TestNG and avoiding thread-safety issues?
- Explain parallel execution in TestNG and configure it. **[x2 exact repeat]**
- Explain Parallel Execution.
- How do you run parallel tests in Selenium?
- Implement parallel test execution
- Explain parallel execution in TestNG and how you implement it.
- How do you make your framework thread-safe for parallel execution?
- How do you execute tests in parallel using TestNG or Maven?
- How you achieve parallel execution using testng?
- TestNG: Parallel executions, Grouping?
- Syntax to perform parallel testing in TestNG and what do you write in <suite tag> also what do you mention in double quotes like parallel = " "
- How do you implement parallel test execution?
- How to run tests in parallel?
- Parallel execution in TestNG
- Running tests in parallel

[UNIQUE] [Section 1 Critical] Q47. How do you execute only the failed test cases (e.g. 50 tests, 5 failed) in TestNG?

- How to execute failed test cases in TestNG?
- How do you avoid running all tests when only one fails in a suite?
- Started automation test suite execution and few test cases are failed in a test run. How can you execute only failed test cases at once (with one click) what design pattern do we use when we trigger different browsers?
- Out of 50 testcases, how you will run only the failed testcases?

[UNIQUE] [Section 1 Critical] Q48. Hard vs soft assertions and Assert vs Verify: when do you use which?

- SoftAssert vs Assert - when to use what
- When to use hard assertions and soft assertions?
- Explain TestNG assertions and how they differ from standard assertions. **[x2 exact repeat]**
- Explain soft assertions and hard assertions in TestNG **[x2 exact repeat]**
- What is the difference between Assert and Verify?
- Difference between assert and verify?
- What is the difference between assertions and verifications?
- Implement soft assertions

[UNIQUE] [Section 2 High] Q110. What are TestNG Listeners (ITestListener, IReporter)? What types exist and how do you implement a custom listener?

- Explain TestNG Listeners and how to implement custom listeners. **[x2 exact repeat]**
- What are TestNG Listeners and how do you use them?
- Explain the role of Listeners ITestListener & IReporter in TestNG
- Explain the purpose of listeners? is it the selenium concept of TestNG?
- Types of Listeners?
- Listeners and ITestListener interface

[UNIQUE] [Section 2 High] Q111. How do you group test cases and run specific groups in TestNG?

- Explain test grouping and how to run specific test groups. **[x2 exact repeat]**
- How do you group test cases?
- What is the purpose of TestNG groups?
- How to group test cases
- How can you group test cases in TestNG?

[UNIQUE] [Section 2 High] Q112. How do you create dependencies between tests (dependsOnMethods)?

- Explain test dependency in TestNG: @Test(dependsOnMethods). **[x2 exact repeat]**
- Dependency between methods
- How do you create dependencies between tests in TestNG?

[UNIQUE] [Section 2 High] Q113. How do you skip a test in TestNG (enabled=false, SkipException)?

- How to skip tests in TestNG? Explain @Test(enabled = false) and SkipException **[x2 exact repeat]**
- How to skip a test in TestNG? **[x2 exact repeat]**
- How to skip a test

[UNIQUE] [Section 2 High] Q114. What are the @Test attributes (invocationCount, timeOut, expectedExceptions, groups, alwaysRun)?

- What is @Test annotation and its attributes? Explain invocationCount, timeOut, expectedExceptions. **[x2 exact repeat]**
- What are attributes of @Test annotation? Explain priority, groups, and alwaysRun. **[x2 exact repeat]**
- What is invocationcount in testng?

[UNIQUE] [Section 2 High] Q115. How do you run the same test method many times (e.g. 100 times with the same data)?

- Case Scenario: How to run the same method 100 times in TestNG with the same data?
- How can you run the same tests for 10 times?

[UNIQUE] [Section 2 High] Q116. How do you generate TestNG reports?

- How to generate TestNG reports and integrate with reporting tools? **[x2 exact repeat]**
- How do you generate HTML reports in TestNG?
- How do you generate reports in TestNG? **[x2 exact repeat]**

[UNIQUE] [Section 3 Medium] Q180. What is the @Factory annotation? Factory vs DataProvider (with a real-time example)?

- What is the use of @Factory annotation in TestNG?
- Difference between @Factory and @DataProvider?
- @Factory explain with real time example?

[UNIQUE] [Section 4 Low] Q199. What are custom annotations in TestNG, and what is the role of annotations in framework optimization? Can test behavior be changed at runtime?

- Custom annotations in TestNG
- What's the role of annotations in framework optimization?
- Is it possible to change the behaviour of a test at runtime?

Cucumber & BDD (77 removed)

[UNIQUE] [Section 1 Critical] Q49. What is BDD and Cucumber? Why is it used with Selenium? Explain Gherkin syntax and keywords. Have you used it?

- Cucumber Framework - Explained BDD concepts and wrote a scenario outline for login functionality along with a runner class implementation.
- What is Cucumber and BDD? Explain Gherkin syntax. **[x2 exact repeat]**
- What is Cucumber, and why is it used with Selenium?
- What is BDD and have you used Cucumber?
- What is Cucumber, and why is it used in BDD?
- What is BDD?
- How do you implement BDD in test automation?
- What is Cucumber? **[x2 exact repeat]**
- What is Gherkin syntax?
- What is Cucumber and why is it used in automation?
- What is Gherkin in Cucumber?
- What are Gherkin keywords, and can you name some of them? (e.g., Given, When, Then, And, But)

[UNIQUE] [Section 1 Critical] **Q50. How are step definitions created and mapped to feature file steps? What if multiple definitions match the same step?**

- Explain step definitions and how to create them. **[x2 exact repeat]**
- How are step definitions mapped to feature file steps?
- What happens if multiple step definitions match the same step?
- Step definition in Cucumber
- How do you create Step Definitions in Cucumber?

[UNIQUE] [Section 1 Critical] **Q51. What are hooks in Cucumber (@Before, @After)? Hooks vs tags.**

- How to connect files in Cucumber? What are hooks?
- What is the use of hooks in Cucumber?
- What are Hooks in Cucumber? Explain @Before and @After. **[x2 exact repeat]**
- Explain the different types of hooks (@Before , @After , etc.) in Cucumber.
- Hooks 3 Before and After
- What are hooks in Cucumber?
- What is a test hook?
- Difference between hooks and tags?

[UNIQUE] [Section 1 Critical] **Q52. What are tags in Cucumber? How do you run specific scenarios or combinations of tags?**

- Explain tags in Cucumber and how to run specific tags. **[x2 exact repeat]**
- Cucumber tags and annotations?
- Cucumber tags? And how to run different combinations of tags when multiple tags are present
- What are tags in Cucumber, and how are they used?
- Can you run specific scenarios based on tags? How?
- Tagging and filtering scenarios
- How can you run specific scenarios or feature files?

[UNIQUE] [Section 1 Critical] **Q53. What is the difference between Scenario and Scenario Outline? Explain Examples.**

- Explain Scenario Outline and Examples in Cucumber **[x2 exact repeat]**
- Difference between Scenario and Scenario Outline?
- What is the difference between Scenario and Scenario Outline ?
- Scenario Outline usage

[UNIQUE] [Section 1 Critical] **Q54. What is the Cucumber runner class and how is it configured (glue, features, plugins)? Write the skeleton.**

- Explain Cucumber runner class and its configuration. **[x2 exact repeat]**
- Write skeleton of test runner?
- What is the role of the Runner class in Cucumber?
- Runner class and glue code
- Cucumber options and annotations
- What is the purpose of the Cucumber Runner Class?
- What is a test runner?

[UNIQUE] [Section 1 Critical] **Q55. How do you integrate Cucumber with Selenium WebDriver and TestNG?**

- How do you integrate Cucumber with TestNG?
- How do you integrate Cucumber with Selenium WebDriver?
- Cucumber with TestNG
- How do you integrate Cucumber with Selenium and TestNG?

[UNIQUE] [Section 2 High] **Q117. What are Background in Cucumber and how do you implement it?**

- How to implement background in Cucumber feature files? **[x2 exact repeat]**
- What is background in Cucumber?
- Background keyword in Cucumber

[UNIQUE] [Section 2 High] **Q118. What are Data Tables? How do you convert them to a list or map? DataTable vs Examples? How do you handle test data in scenarios?**

- What are Data Tables in Cucumber and how to use them? **[x2 exact repeat]**
- What is the difference between DataTable and Examples ?
- How do you convert a DataTable to a list or map in step definitions?
- Data tables in Cucumber
- How do you handle Data Tables in Cucumber?
- How do you handle test data in Cucumber scenarios?

[UNIQUE] [Section 2 High] **Q119. How do you pass parameters in step definitions (regex, placeholders)?**

- Explain different ways to pass parameters in Cucumber step definitions. **[x2 exact repeat]**
- How do you parameterize tests in Cucumber?
- How do you parameterize steps in Cucumber? (Explain usage of regular expressions or <placeholders> in steps.)

[UNIQUE] [Section 2 High] **Q120. How do you run Cucumber tests in parallel?**

- How to implement parallel execution in Cucumber? **[x2 exact repeat]**
- How do you run Cucumber tests in parallel?
- Can you run Cucumber tests in parallel? How?

[UNIQUE] [Section 2 High] **Q121. How do you share data between steps (World/context object)?**

- Explain World context/context object in Cucumber. **[x2 exact repeat]**
- How do you share data between steps in Cucumber?

[UNIQUE] [Section 2 High] **Q122. How do you generate Cucumber reports (HTML)?**

- Explain Cucumber reports and how to integrate with reporting tools. **[x2 exact repeat]**
- Generating Cucumber HTML reports

[UNIQUE] [Section 2 High] **Q123. How do you structure a Cucumber project and feature files?**

- Explain the structure of a Cucumber project.
- How do you structure feature files in Cucumber?
- Structure of a feature file
- What are the main components of a Cucumber framework?
- What does your Cucumber framework structure look like?

[UNIQUE] [Section 2 High] **Q124. How do you write reusable step definitions and follow best practices?**

- Explain step definition best practices and how to create maintainable step definitions **[x2 exact repeat]**
- How do you manage reusable step definitions across multiple feature files?
- Reusable steps in Cucumber

[UNIQUE] [Section 3 Medium] **Q181. What are common challenges in writing Cucumber tests and how do you keep scenarios readable and maintainable?**

- What are some common challenges in writing Cucumber tests, and how do you solve them?
- How do you ensure Cucumber scenarios remain readable and maintainable?

[UNIQUE] [Section 3 Medium] **Q183. Can you share an example of a complex scenario you wrote? How do you ensure collaboration between developers, testers and business in BDD?**

- Can you share an example of how you wrote a complex scenario in Cucumber?
- How do you ensure collaboration between developers, testers, and business stakeholders in BDD?

[UNIQUE] [Section 3 Medium] **Q184. How do you skip scenarios in Cucumber? How do you integrate Cucumber in a Maven project?**

- How to skip scenarios in Cucumber? **[x2 exact repeat]**
- Integrating Cucumber in Maven project

Test Data, Configuration, Reporting & Logging (45 removed)

[UNIQUE] [Section 1 Critical] Q56. How do you implement data-driven testing and manage test data in your framework (Excel, CSV, JSON)?

- DataProvider with JSON + Excel test data
- How do you implement data-driven testing using Excel / CSV / JSON files?
- How to pass test data to tests?
- How do you implement Data-Driven Testing in Selenium Java?
- How do you handle test data in your framework?
- Implement data-driven testing for multiple input combinations.
- What is data-driven testing?
- How is data-driven testing implemented?
- Data handling via Excel/JSON
- Explain Data Driven testing in TestNG.
- How do you implement Data-Driven Testing in TestNG? **[x2 exact repeat]**
- How do you manage a set of Data Tables in Selenium?

[UNIQUE] [Section 1 Critical] Q57. How do you read data from an Excel file in Java? Write the code.

- How do you read data from an Excel file in Java Selenium?
- How do you read excel in the script? (very careful while answering. the counter-question will come as per your answer)
- Write the code to read the value from the excel sheet.
- How to read test data from Excel in Selenium Framework?

[UNIQUE] [Section 1 Critical] Q58. How do you manage environment configuration (URLs, credentials, toggles for DEV/QA/UAT/PROD) and config.properties?

- How do you manage test data and environment configuration in Selenium automation? **[x2 exact repeat]**
- How to implement test environment management?
- Design a config.properties management class
- Implement environment-based test execution
- Manage URLs, credentials, and toggles per environment (DEV/QA/UAT/PROD).
- How do you implement proper configuration management in your framework?
- How do you handle test environment configurations?
- How do you maintain cross-environment configurations?
- Config file management

[UNIQUE] [Section 1 Critical] Q59. How do you generate reports (Extent/Allure) with screenshots, steps and logs? Where are reports stored or published?

- Where automation reports are stored or published
- How do you generate Extent/Allure reports?
- How to implement reporting with screenshots and logs?
- What is Extent Reports, and how do you use it in Selenium?
- How do you implement reporting in your framework?
- Adopt rich reports with screenshots, steps, and categorization.
- What is the reporting tool in your framework? and why?
- How do you handle reports in Cucumber (e.g., Allure or ExtentReports)?
- What is a test execution report and what should it include?
- How do you handle reporting in test frameworks?
- How do you dynamically create test reports after every run?
- Test reports (Extent/Allure)
- How do you generate reports in Selenium (ExtentReports/Allure)?
- How to generate reports?

[UNIQUE] [Section 2 High] Q128. How do you ensure test data independence, isolate data for parallel runs and clean up after tests?

- How to handle test data management and test data cleanup?
- How do you ensure test data independence?
- What is test data management and how should you implement it?
- How do you isolate test data for parallel test execution?
- Why is test data management important in automation?
- How do you clean up after test runs?

[UNIQUE] [Section 2 High] Q129. How do you manage sensitive data like passwords and test credentials?

- How do you manage sensitive data like passwords in your framework?
- What is the best practice for managing test credentials?

[UNIQUE] [Section 2 High] Q130. How do you implement logging in your framework (Log4j)?

- How do you implement logging in your framework?
- How do you implement proper logging in your test framework?
- Logging using Log4j

Cross-browser, Grid, Docker, CI/CD, Maven & Git (31 removed)

[UNIQUE] [Section 1 Critical] Q60. How do you implement cross-browser testing and run across browsers, versions and environments?

- How do you manage test execution across multiple browsers and environments (QA, UAT, Staging)?
- How do you implement cross-browser testing?
- How do you run Selenium tests in multiple browsers?
- How do you run tests in multiple browsers using TestNG?
- Need to validate the app across Chrome / Firefox / Edge and multiple versions.
- How do you handle cross-browser testing in your framework? **[x2 exact repeat]**
- How do you test across multiple browsers using WebDriver?

[UNIQUE] [Section 1 Critical] Q61. Explain CI/CD and how you integrate automation with Jenkins or Azure DevOps.

- Explain CI/CD integration with test automation frameworks.
- How do you integrate Selenium tests with Jenkins?
- What is CI/CD, and why is it important in automation?
- What is continuous integration (CI/CD) and how do you integrate tests?
- How to integrate Selenium with CI/CD (both Jenkins and ADO)?

[UNIQUE] [Section 2 High] Q131. What is Selenium Grid and RemoteWebDriver? How do you run tests on a Grid or remote machine?

- What are Selenium Grid and parallel execution?
- How do you use Selenium Grid?
- What is Selenium Grid?
- How do you run Selenium tests on a Selenium Grid?
- How to run Selenium on remote machine?
- What is RemoteWebDriver and when do you use it?

[UNIQUE] [Section 2 High] Q132. How do you run Selenium tests in Docker or without a local browser?

- How do you run Selenium tests in Docker?
- How do you run Selenium tests without installing a browser locally?
- How to run Selenium in Docker?

[UNIQUE] [Section 2 High] Q133. What is headless browser testing? Advantages, disadvantages and how to run it.

- Explain the headless browser testing and its advantages/disadvantages. **[x2 exact repeat]**
- How do you run Selenium tests in headless mode?
- What is Headless browser testing?
- How do you execute using headless mode?
- Headless browser testing?
- What is headless browser testing and when is it useful?
- How to run tests in headless mode?

[UNIQUE] [Section 2 High] Q134. What is Maven and how is it used? Structure of pom.xml, group id vs artifact id, running TestNG with Maven.

- What is Maven, and how is it used in Selenium projects?
- What is Maven and why do you use it in automation projects?
- Explain the structure of pom.xml file.
- How do you execute TestNG tests using Maven?
- What is difference between group id and artifact id?

[UNIQUE] [Section 2 High] Q135. How do you use Git and version control in your automation project?

- How do you use Git in Selenium projects?
- How do you version control your automation framework?
- Version control integration

Scenario-Based & Real-World Automation (22 removed)

[UNIQUE] [Section 1 Critical] Q62. How would you automate login (including multiple user roles and permissions) and handle authentication?

- How would you automate login functionality for a website? **[x2 exact repeat]**
- Login automation using POM
- Automate login for multiple user roles and validate permissions.
- How do you handle authentication in automation testing?

[UNIQUE] [Section 1 Critical] Q63. How do you decide what to automate and what not to automate? (e.g. a long-running scenario: manual or automation?)

- If there is a scenario that takes a long time to execute, would you prefer Manual Testing or Automation Testing?
- In your Selenium project, how do you decide what to automate and what not to automate?
- How do you decide whether a scenario should be automated or not?
- If there is a scenario that takes a long time to execute, would you prefer Manual Testing or Automation Testing? Why?

[UNIQUE] [Section 2 High] Q137. How do you handle CAPTCHA in automation?

- Scenario: How do you handle captcha in automation?
- How do you automate CAPTCHA in Selenium?
- How do you handle CAPTCHA in automation?
- Can Selenium handle CAPTCHAs?
- How do you automate CAPTCHA-based logins?

[UNIQUE] [Section 2 High] Q138. How do you handle OTP in automation?

- Handling OTPs in automation?
- How do you test an OTP-based login system?
- Automating flows that include CAPTCHA/OTP.

[UNIQUE] [Section 2 High] Q139. You are asked to automate a functionality that is not fully developed. How do you handle it?

- You are asked to automate a functionality that is not yet fully developed. How do you handle this? **[x2 exact repeat]**

[UNIQUE] [Section 3 Medium] Q185. You are asked to automate a legacy application. What is your approach?

- You've been asked to automate a legacy application. What is your approach? **[x2 exact repeat]**

[UNIQUE] [Section 3 Medium] Q186. How would you design tests for a shopping cart and a multi-step form with validation?

- How would you design tests for a shopping cart functionality?
- How would you test a multi-step form with validation?

[UNIQUE] [Section 3 Medium] Q187. How do you validate an email sent after registration or an email verification link?

- How would you automate a Test scenario where you need to check if an email is sent after a user registration? **[x2 exact repeat]**
- How do you validate email verification links sent during signup?

[UNIQUE] [Section 3 Medium] Q188. How do you test responsive layouts and mobile web with Selenium?

- How do you test responsive design in Selenium?
- Verify responsive behavior on common breakpoints.
- How do you test responsive layouts in Selenium?
- How do you test mobile web applications using Selenium?

[UNIQUE] [Section 3 Medium] Q189. How do you measure page load time through automation?

- How do you measure page load time in Selenium?
- How do you check loading times via automation?

Java Concepts in the Selenium Framework (18 removed)

[UNIQUE] [Section 1 Critical] Q64. Explain OOP concepts and where you applied them in your framework (inheritance, polymorphism, abstraction, encapsulation).

- OOP concepts and where you applied them in a framework
- Explain the main OOP concepts and how they apply in Selenium.
- Explain abstraction and interface respect of selenium with some example
- Have you used interface in your framework other than selenium interfaces?
- How do you achieve inheritance in your framework?
- Where do you used inheritance in selenium?

[UNIQUE] [Section 1 Critical] Q65. Overloading vs overriding in a Selenium framework, with examples.

- Overloading vs Overriding in Selenium framework?
- Explain method overloading and overriding in Selenium context.
- Explain method overloading with selenium and some example

[UNIQUE] [Section 2 High] Q140. How do you use interfaces and abstract classes in Selenium? Why are interfaces important in automation?

- How do you use Interfaces in Selenium?
- How do you use Abstract Classes in Selenium?
- Explain the concept of Interfaces in Java and why they're important in Automation.

[UNIQUE] [Section 2 High] Q141. Which Java collections have you used in your framework? Do you use a property file, and which Java concept does it use?

- Which collections have you used in your framework?
- What are Java Collections, and where do you use them in Selenium?
- What are Collections in Java? Which ones do you use most in automation?
- Do you use the property file in your framework? If yes, then which java concept gets utilize here? (Java Collection)

[UNIQUE] [Section 2 High] Q142. How do you handle exceptions in Java Selenium scripts? Checked vs unchecked exceptions.

- How do you handle exceptions in Java Selenium scripts?
- What is the difference between Checked and Unchecked exceptions?
- Explain Exception Handling in Java and how you use it in automation.

[UNIQUE] [Section 2 High] Q143. How are constructors, the this and super keywords, access modifiers and multiple classes used in the framework?

- What are Access Modifiers in Java, and why are they important in Selenium framework design?
- What is a Constructor in Java, and how is it used in Selenium POM?
- How do you handle multiple classes in Selenium automation?
- What is this keyword in Java, and where do you use it in Selenium?
- How do you use super keyword in Selenium?

Automation Strategy, Metrics & Career (16 removed)

[UNIQUE] [Section 2 High] Q144. What metrics do you track in test automation? How do you track coverage and calculate ROI?

- What are the key metrics to track for your test automation framework?
- What metrics do you track in test automation?
- How do you track test coverage in automation?
- How do you calculate ROI of automation?

[UNIQUE] [Section 2 High] Q146. Describe a time you owned the framework during a critical release. What are your key achievements in automation?

- Ownership -> Time you maintained framework during critical release
- Can tell about any achievements you have done in automation?

[UNIQUE] [Section 3 Medium] Q190. What is the risk of over-automation? How do you avoid duplicate and redundant tests?

- What's the risk of over-automation?
- How do you avoid test duplication?
- How do you identify redundant test cases?

[UNIQUE] [Section 3 Medium] Q191. Why avoid hardcoded values, and how do you ensure reusability in test methods?

- Why should you avoid hardcoded values in test scripts?
- How do you ensure reusability in test methods?

[UNIQUE] [Section 3 Medium] Q192. How should you handle test dependencies and execution order? Why should tests be atomic?

- How should you handle test dependencies and execution order?
- What is test dependency and how to avoid it?
- Why should tests be atomic?

[UNIQUE] [Section 4 Low] Q202. How do you conduct automation code reviews and walkthroughs, document the framework and version test cases?

- How do you perform Automation Code Review/ Walk Through in your project?
- How do you document your automation framework?
- How do you implement test case versioning?

[UNIQUE] [Section 4 Low] Q204. What are your daily activities as an automation engineer? How do you stay current and balance manual and automation work?

- What are your daily activities as an Automation Engineer?
- How do you stay current with new automation trends?
- How do you balance manual and automation tasks?

[UNIQUE] [Section 4 Low] **Q205. How do you test the stability, data reusability, performance and rollback of the automation framework?**

- How do you test automation framework stability?
- How do you test if automation test data is re-usable?
- How do you test performance of automation framework?
- How do you test rollback in automation framework if test fails?

Selenium Coding Questions (2 removed)

Source file: Selenium Coding Questions. Original 23 questions, 21 unique kept, 2 removed. Prioritized PDF: Selenium_Coding_Questions_Prioritized.pdf.

Test Data, Config & API Data Handling (1 removed)

[UNIQUE] [Section 1 Critical] Q3. Read properties from a config file (read key-value pairs from a properties file).

- Read properties from a config file
- How do you read key-value pairs from a properties file?

Collections & Streams on Test Results (1 removed)

[UNIQUE] [Section 2 High] Q16. How do you use a Queue in automation scenarios (e.g. simulate a test queue for parallel execution)?

- How do you use Queue in automation scenarios?
- How do you simulate a test queue for parallel execution?

SQL Interview Questions (22 removed)

Source file: SQL Interview Questions. Original 97 questions, 75 unique kept, 22 removed. Prioritized PDF: SQL_Interview_Questions_Prioritized.pdf.

Database Concepts & Keys (1 removed)

[UNIQUE] [Section 1 Critical] Q2. What is the difference between DELETE, TRUNCATE and DROP?

- What is the difference between DELETE, TRUNCATE, and DROP?
- Difference between DELETE,DROP & TRUNCATE?

Joins & Set Operations (5 removed)

[UNIQUE] [Section 1 Critical] Q6. What are joins and their types? Explain INNER, LEFT, RIGHT, FULL and the difference between inner and outer join.

- What are joins and their types?
- What is the difference between INNER JOIN, LEFT JOIN, RIGHT JOIN, and FULL JOIN?
- Explain inner join and outer join in SQL?

[UNIQUE] [Section 1 Critical] Q7. How do you find records with no matching rows (employees without orders or department, products never sold, rows in one table but not in another)?

- How do you find employees who don’t have any orders?
- SQL Query - Find employees without department (left join usage)
- SQL Query - Find products that were never sold.
- How do you find records present in one table but not in another?

Subqueries, Top-N & Ranking (5 removed)

[UNIQUE] [Section 1 Critical] Q8. Write a query to find the second highest salary (also: Nth highest).

- Write query to find 2nd highest salary.
- How do you find the second highest salary from a table?
- SQL Query - Retrieve the second highest salary from the employee table

[UNIQUE] [Section 1 Critical] Q9. Retrieve the top N records (top 5 records, top 3 highest paid employees, salaries within the top 3 in the company).

- SQL Query - Get the top 3 highest paid employees
- How do you find employees whose salary is within the top 3 highest salaries in the company?
- How do you retrieve the top 5 records from a table?

[UNIQUE] [Section 2 High] Q26. Find customers with orders above the average order value, or who placed more orders than the average per customer.

- SQL Query - Retrieve customers with orders above the average order value
- SQL Query - Find customers who ordered more than the average number of order per customer.

Aggregation & GROUP BY Queries (3 removed)

[UNIQUE] [Section 1 Critical] Q11. Find the total salary and the highest salary in each department.

- How do you find the total salary for each department?
- How do you find the highest salary in each department?

[UNIQUE] [Section 1 Critical] Q12. Show the count of orders per customer and the total revenue per product.

- SQL Query - Show the count of orders per customer
- SQL Query - Calculate the total revenue per product

[UNIQUE] [Section 2 High] Q31. Get the total revenue and number of orders per region, and monthly sales revenue with order count.

- SQL Query - Get the total revenue and the number of orders per region.
- SQL Query - Get monthly sales revenue and order count.

Duplicates & Data Comparison (1 removed)

[UNIQUE] [Section 1 Critical] Q14. How do you find duplicate records in a table?

- How do you find duplicate records in a table?
- SQL Query - Find duplicate records in a table

DML, NULLs, Functions & Date/String Handling (4 removed)

[UNIQUE] [Section 2 High] Q42. How do you check for NULL values and replace them with a default value (IS NULL, COALESCE/IFNULL)?

- How do you check for NULL values in SQL?
- How do you replace NULL values with a default value?

[UNIQUE] [Section 2 High] Q47. How do you get the difference between two dates and find employees hired in the last 30 days?

- How do you get the difference between two dates in SQL?
- How do you find all employees hired in the last 30 days?

[UNIQUE] [Section 3 Medium] Q56. How do you retrieve all employees who joined in a given year (2023) or on weekends?

- SQL Query - Retrieve all employees who joined in 2023
- SQL Query - Find the employees hired on weekends.

[UNIQUE] [Section 3 Medium] Q60. How do you convert a string to uppercase and find its length?

- How do you convert a string to uppercase in SQL?
- How do you find the length of a string in SQL?

Business Scenario Queries (Orders, Customers, Revenue) (2 removed)

[UNIQUE] [Section 2 High] Q48. Get the latest (last) order placed by each customer, along with the order amount; also the first and last order date.

- SQL Query - Get the latest order placed by each customer.
- SQL Query - Show last purchase for each customer along with order amount.
- SQL Query - Identify the first and last order date for each customer.

Advanced Analytics & Window Functions (1 removed)

[UNIQUE] [Section 4 Low] Q75. Detect customers above their historical 90th percentile purchase amount, or with revenue below the 10th percentile.

- SQL Query - Detect customers whose purchase amount is higher than their historical 90th percentile.
- SQL Query - Identify customers with revenue below the 10th percentile.

API Manual Testing Questions (54 removed)

Source file: API Manual Testing Questions. Original 137 questions, 83 unique kept, 54 removed. Prioritized PDF: API_Manual_Testing_Questions_Prioritized.pdf.

API Testing Fundamentals (3 removed)

[UNIQUE] [Section 1 Critical] Q1. What is API testing and why is it important?

- What is API testing and why is it important?
- What is API testing?
- Why is API testing important?

[UNIQUE] [Section 2 High] Q23. What is an API and how does it work? What are the different types of APIs?

- What is an API and how does it work?
- What are the different types of APIs?

REST, SOAP & HTTP Concepts (22 removed)

[UNIQUE] [Section 1 Critical] Q3. Explain the HTTP methods used in API testing (GET, POST, PUT, PATCH, DELETE).

- Explain HTTP methods in API testing.
- Method in REST.
- Common HTTP methods used in REST?

[UNIQUE] [Section 1 Critical] Q4. What is the difference between PUT and PATCH?

- What is the difference between PUT and PATCH?
- Difference between PUT and PATCH? **[x2 exact repeat]**
- Difference between PUT and PATCH call

[UNIQUE] [Section 1 Critical] Q5. What is idempotency? Which HTTP methods are idempotent?

- What is idempotency in HTTP methods?
- What is idempotency in REST APIs?
- What are idempotent methods?

[UNIQUE] [Section 1 Critical] Q6. Explain HTTP status codes, their categories and the ones you commonly use.

- Explain different HTTP status codes and their meanings.
- Categories of HTTP status codes.
- Explain HTTP status codes commonly used in API testing
- What are the API status codes, you have come across?
- Common HTTP status codes?

[UNIQUE] [Section 1 Critical] Q8. What is the difference between SOAP and REST APIs? How do you test SOAP APIs?

- What's the difference between SOAP and REST APIs?
- Explain SOAP API testing and how it differs from REST.
- What is the difference between SOAP and REST APIs?
- What are the key differences between REST and SOAP?
- Difference between REST and SOAPUI.
- Difference between REST and SOAP?

[UNIQUE] [Section 2 High] Q26. What are headers, body and parameters in requests and responses?

- What are headers in API requests and responses?
- What are Headers, Body, and Parameters?

[UNIQUE] [Section 2 High] Q27. How do you handle query parameters and path parameters?

- How do you pass query parameters in Postman?
- How to handle query parameters and path parameters?
- How do you pass query parameters and headers?

[UNIQUE] [Section 2 High] Q28. What is the difference between URI, URL and endpoint?

- Difference between URI, URL, and Endpoint.
- Difference between URI and Endpoint?

[UNIQUE] [Section 2 High] Q29. What is JSON and why is it preferred over XML?

- What is JSON and why is it preferred over XML?
- What is JSON and why is it used?

[UNIQUE] [Section 2 High] Q30. How do you test APIs with different content types (JSON, XML, form data)? Raw vs form-data body?

- How to test API with different content types (JSON, XML, Form Data)?
- Difference between raw and form-data body?

Postman: Usage, Scripting & Chaining (9 removed)

[UNIQUE] [Section 1 Critical] Q9. What is Postman and why is it widely used in API testing?

- What is Postman and why is it widely used in API testing?
- What is Postman and why is it used?

[UNIQUE] [Section 1 Critical] Q10. What are Collections and Environments in Postman? How do you import/export them?

- What are Postman Collections and why are they important?
- What are environments in Postman?
- What are Collections and Environments?
- How do you import/export collections and environments in Postman?

[UNIQUE] [Section 1 Critical] Q12. How do you chain requests: capture a value from one response and pass it to another request?

- How do you capture specific responses value and pass to other request?
- How you get the response from one api and send to another api?
- How do you chain multiple requests?

[UNIQUE] [Section 1 Critical] Q14. How do you automate API tests in Postman and validate responses with scripts?

- How do you automate API tests in Postman?
- How do you validate responses in Postman?

[UNIQUE] [Section 1 Critical] Q15. How do you implement data-driven API testing (Postman data files, parameterization)?

- How do you test APIs in Postman with data-driven approach?
- Explain parameterization and data-driven testing in REST API tests.
- How to implement data-driven testing for APIs?

Response Validation & Test Design (5 removed)

[UNIQUE] [Section 1 Critical] Q16. How do you validate an API response (status code, body, headers, response time)? What are the most common validations?

- How do you validate an API response?
- What are the most common API testing validations?

[UNIQUE] [Section 1 Critical] Q17. How do you validate the JSON schema? What is schema validation and how do you test if a response schema changes?

- How to validate JSON schema in REST API tests?
- What is schema validation and why is it important?
- How do you test if an API response schema changes?

[UNIQUE] [Section 1 Critical] Q18. How do you test API error handling and error codes?

- How do you test API error handling?
- How do you test API error codes?

[UNIQUE] [Section 2 High] Q34. How do you handle timeouts, retries and error scenarios in API testing?

- How to handle timeouts, retries, and error scenarios in API testing?
- How do you test API timeout handling?

Authentication, Security & Rate Limiting (6 removed)

[UNIQUE] [Section 1 Critical] Q19. Explain authentication types (Basic, OAuth, Bearer token). How do you handle authentication in Postman?

- Explain authentication types (Basic, OAuth, Bearer token).
- How do you send authentication credentials in Postman?
- How do you handle authentication in Postman?

[UNIQUE] [Section 2 High] Q39. How do you test API authentication (OAuth, JWT)? How do you handle tokens, OAuth and CORS in Postman?

- How do you test API authentication (OAuth, JWT)?
- How do you handle authentication tokens, OAuth, and CORS issues while performing API manual testing using tools like Postman?

[UNIQUE] [Section 2 High] Q41. How do you test API security (in Postman)?

- How do you test API security in Postman?
- How do you handle API security testing in Postman?

[UNIQUE] [Section 2 High] Q42. How do you test API rate limiting (throttling)?

- How do you test API rate limiting?
- How do you test API rate limiting (throttling)?
- How to test rate limiting?

Scenario-Based API Testing (7 removed)

[UNIQUE] [Section 1 Critical] Q21. How do you approach testing a new API when no documentation is available, or the requirements are incomplete and the behaviour changes often?

- How do you approach testing a new API when no documentation is available?
- A major new API endpoint has been delivered by development, but the requirements are incomplete and the endpoint behaviour changes every few days. How would you approach validation of this endpoint using automation (and manual) methods?

[UNIQUE] [Section 2 High] Q43. How do you test API pagination?

- How do you test API pagination? **[x2 exact repeat]**
- How to test pagination in APIs?

[UNIQUE] [Section 2 High] Q45. How do you test APIs that return asynchronous responses?

- How do you test APIs that return asynchronous responses?
- How do you test APIs that return asynchronous results?

[UNIQUE] [Section 2 High] Q46. How do you test APIs that depend on other services (and test APIs in a microservices architecture)?

- How do you test APIs that have dependencies on other services?
- How do you test APIs in a microservices architecture?

[UNIQUE] [Section 2 High] **Q48. How do you validate API caching behaviour?**

- How do you validate API caching behavior?
- How do you test if API caching is implemented correctly?

[UNIQUE] [Section 2 High] **Q51. How do you test the performance of an API (Postman, product search API)?**

- How do you test an API for performance using Postman?
- How do you test performance of product search API?

API Contract Testing (2 removed)

[UNIQUE] [Section 2 High] **Q58. What is API contract testing and why is it important?**

- What is API contract testing?
- Why is API Contract Testing important?

[UNIQUE] [Section 2 High] **Q59. What tools are used for API contract testing? What is Pact and how does it work?**

- What are some tools commonly used for API Contract Testing?
- What is Pact, and how does it work in API Contract Testing?

API Automation Questions (178 removed)

Source file: API Automation Questions. Original 299 questions, 121 unique kept, 178 removed. Prioritized PDF: API_Automation_Questions_Prioritized.pdf.

API Testing & HTTP Fundamentals (8 removed)

[UNIQUE] [Section 1 Critical] Q2. Explain the HTTP methods used in RESTful APIs (GET, POST, PUT, PATCH, DELETE).

- Explain HTTP methods in API testing.
- What are the main HTTP methods used in RESTful APIs?

[UNIQUE] [Section 1 Critical] Q4. Explain HTTP status code categories and what 200, 201, 400, 401, 404 and 500 mean.

- Categories of HTTP status codes.
- What does the status code 200 mean?
- What does the status code 201 mean?
- What does the status code 400 mean?
- What does the status code 401 mean?
- What does the status code 404 mean?
- What does the status code 500 mean?

[UNIQUE] [Section 1 Critical] Q5. What is the difference between SOAP and REST APIs? How do you test SOAP?

- Explain SOAP API testing and how it differs from REST.
- What is the difference between SOAP and REST APIs?

REST Assured: Requests, Specifications & Parameters (36 removed)

[UNIQUE] [Section 1 Critical] Q7. What is REST Assured? Explain its structure and components, and how you set it up in a project.

- Explain RestAssured library and its basic structure.
- Explain components of REST API / Rest Assured.
- What is REST Assured? **[x3 exact repeat]**
- How do you set up REST Assured in a project?

[UNIQUE] [Section 1 Critical] Q8. Write a basic GET request using REST Assured.

- Write a basic GET request using REST Assured
- How do you perform a GET request using REST Assured?
- How to perform a GET request
- How to send GET and POST requests?

[UNIQUE] [Section 1 Critical] Q9. How do you send a POST request with a JSON body using REST Assured?

- How to perform POST request with request body?
- How do you send a POST request with a JSON body using REST Assured?
- How to send POST request with payload

[UNIQUE] [Section 1 Critical] Q10. What are Request Specification and Response Specification? How do you create and reuse them (base URI)?

- Difference between Request Specification and Response Specification.
- Explain RequestSpecification and ResponseSpecification
- What is a request specification in REST Assured?
- How do you create and use a request specification in REST Assured?
- What is a response specification in REST Assured?
- How do you create and use a response specification in REST Assured?
- How to set Base URI and Request Specification?
- How to reuse request specifications?
- How do you reuse request and response specifications across multiple tests in REST Assured?

[UNIQUE] [Section 1 Critical] Q11. How do you handle query parameters and path parameters in REST Assured?

- How to handle query parameters and path parameters?
- How do you add query parameters to a request in REST Assured?
- How do you handle path parameters in REST Assured?
- Using path and query parameters
- How to pass query or path parameters?

[UNIQUE] [Section 1 Critical] Q12. How do you log request and response details (to console or a file)?

- How to handle API request/response logging in RestAssured?
- How to log request and response details? **[x2 exact repeat]**
- How do you log request and response details in REST Assured?
- How do you capture request and response logs to a file in REST Assured?
- Logging requests and responses

[UNIQUE] [Section 2 High] Q39. How do you perform PUT (with validation) and DELETE requests in REST Assured? Write sample code.

- How do you perform PUT validations in API automation testing using Rest Assured? Write sample code.
- How do you send a PUT request with REST Assured?
- How do you validate a DELETE request response using REST Assured?

[UNIQUE] [Section 2 High] Q40. How do you set headers and cookies, and extract cookies from a response?

- How do you add headers to a request in REST Assured?
- How do you send cookies with a request in REST Assured?
- Setting headers and cookies
- How do you extract cookies from a response in REST Assured?

[UNIQUE] [Section 2 High] Q41. How do you test APIs with different content types (JSON, XML, form data)?

- How to test API with different content types (JSON, XML, Form Data)?
- How do you handle different content types in REST Assured?

[UNIQUE] [Section 2 High] Q42. How do you perform file upload and multipart form-data requests?

- How to handle file uploads in API testing?
- How do you perform file upload in REST Assured?
- How do you perform multipart form data requests in REST Assured?
- How do you test APIs that require multipart file uploads using REST Assured?

REST Assured: Response Validation & JSON Handling (39 removed)

[UNIQUE] [Section 1 Critical] Q13. How do you validate an API response? What are the most common validations?

- How do you validate an API response?
- What are the most common API testing validations?

[UNIQUE] [Section 1 Critical] Q14. How do you validate the status code and response body with assertions (and multiple assertions in one request)?

- How do you validate the status code of a response using REST Assured?
- How do you validate the response of a POST request with a payload in REST Assured?
- How do you use REST Assured to validate the status code and headers of a response?
- How do you validate multiple assertions in a single REST Assured request?
- Validating status code and response body
- Assertions in Rest Assured
- How to validate status and response body?

[UNIQUE] [Section 1 Critical] Q15. How do you validate JSON response data, extract values and use JsonPath (including checking a key is present)?

- How do you validate a JSON response in API automation testing using Java?
- API Testing in Java - Wrote code to validate request and response payloads using REST-assured.
- How to extract response values in REST Assured?
- Handle dynamic JSON responses with JsonPath
- How do you validate JSON response data using REST Assured?
- How do you extract a value from the JSON response using REST Assured?
- How do you validate that a key is present in the JSON response using REST Assured?
- Response JSON parsing
- How to extract data from JSON response?
- How do you validate JSON responses?

[UNIQUE] [Section 1 Critical] Q16. How do you validate JSON schema in REST Assured? What is schema validation and why is it important?

- How do you validate JSON schema in Rest Assured? **[x2 exact repeat]**
- What is schema validation and why is it important?
- How do you test if an API response schema changes?
- How to validate JSON schema in REST Assured?
- JSON schema validation
- What is schema validation?

[UNIQUE] [Section 1 Critical] Q17. How do you validate response time?

- How to validate response time in API automation?
- How do you validate the response time of an API request using REST Assured?
- How do you measure response times in REST Assured?
- How to validate response time?

[UNIQUE] [Section 1 Critical] Q18. How do you test error handling, error responses, error codes and negative scenarios?

- How do you handle API response codes in automation framework?
- How do you test API error handling?
- How do you test API error codes?
- How do you test for error responses in REST Assured?
- How to test negative scenarios?
- What is error handling in APIs?

[UNIQUE] [Section 2 High] Q43. How do you validate response headers, cookies and content type?

- How do you validate the response content type using REST Assured?
- How do you validate response headers using REST Assured?
- How to validate headers and cookies?

[UNIQUE] [Section 2 High] Q44. How do you validate JSON arrays and lists containing specific values?

- How do you validate a JSON array in the response using REST Assured?
- How do you validate that a list in a JSON response contains specific values using REST Assured?

[UNIQUE] [Section 2 High] Q45. How do you handle nested and deeply nested JSON responses?

- How do you handle nested JSON responses in REST Assured?
- How do you validate deeply nested JSON structures in REST Assured?
- How do you validate complex JSON responses that contain lists and objects in REST Assured?
- How to verify nested JSON objects?

[UNIQUE] [Section 2 High] Q46. How do you handle dynamic data and dynamic payloads or JSON files?

- How do you handle dynamic data in API tests using REST Assured?
- How to handle dynamic payloads or JSON files?

[UNIQUE] [Section 3 Medium] Q96. How do you parse XML responses and validate an XML schema?

- Explain JSON and XML response parsing in RestAssured.
- How do you parse XML responses using REST Assured?
- How do you validate XML schema in REST Assured?

Authentication, Security & Rate Limiting (17 removed)

[UNIQUE] [Section 1 Critical] Q19. How do you handle authentication in REST Assured (basic, OAuth2, token-based, dynamic tokens)?

- How do you automate APIs that require authentication tokens or OAuth, and how do you handle related configurations during automation execution?
- How to handle request/response headers and authentication in RestAssured?
- How to handle authentication in REST Assured?
- How do you handle basic authentication in REST Assured? **[x2 exact repeat]**
- How do you handle OAuth2 authentication in REST Assured?
- Token-based authentication
- How to handle authentication?
- How to handle dynamic tokens?

[UNIQUE] [Section 2 High] Q49. How do you test API authentication (OAuth, JWT)? How do you validate JWT tokens?

- How do you test API authentication (OAuth, JWT)?
- How do you validate JWT tokens in REST Assured?

[UNIQUE] [Section 2 High] Q50. What is API security testing? How do you test broken authentication, SQL injection and XSS on an API?

- What is API security testing?
- How do you perform SQL injection testing on an API using REST Assured?
- How do you perform cross-site scripting (XSS) testing on an API using REST Assured?
- How do you test for broken authentication vulnerabilities in an API?
- How do you test API security in Postman?
- How do you handle API security testing in Postman?

[UNIQUE] [Section 2 High] Q51. How do you test API rate limiting (throttling) and handle it in automation?

- How do you test API rate limiting?
- How do you test API rate limiting (throttling)?
- How to test rate limiting?
- How do you handle rate limiting in API tests using REST Assured?

API Automation Framework Design (21 removed)

[UNIQUE] [Section 1 Critical] Q23. Write a POJO class for serialization/deserialization. How do you use Jackson/Gson and POJOs in API tests?

- Write a POJO class for serialization/deserialization
- How to use Jackson/Gson for JSON parsing?
- Using POJOs in API tests

[UNIQUE] [Section 1 Critical] **Q24. How do you implement data-driven testing for APIs (TestNG DataProvider, JUnit, Excel/JSON)?**

- Explain parameterization and data-driven testing in REST API tests.
- How to implement data-driven testing for APIs?
- What is data-driven testing?
- How do you implement data-driven testing in REST Assured using TestNG?
- How do you implement data-driven testing in REST Assured using JUnit?
- Data-driven API testing
- How to perform data-driven testing?

[UNIQUE] [Section 1 Critical] **Q25. How do you integrate REST Assured with TestNG or JUnit?**

- Integration of Rest Assured with TestNG
- How to integrate with TestNG or JUnit?

[UNIQUE] [Section 1 Critical] **Q26. How do you implement request chaining (extract a value and pass it to the next request)?**

- Implement chaining of API requests
- Request chaining

[UNIQUE] [Section 2 High] **Q52. Design a config.properties management class and manage environment-based execution (dev, staging, prod).**

- Design a config.properties management class
- Implement environment-based test execution
- How do you manage different environments (dev, staging, prod) in REST Assured?
- How to parameterize environment/config values?

[UNIQUE] [Section 2 High] **Q54. Create a response validator utility and a reusable API helper class.**

- Create a response validator utility
- Create a reusable API helper class

[UNIQUE] [Section 2 High] **Q55. Design endpoint management using Enums. Design a page object model for API endpoints.**

- Design endpoint management using Enums
- Design page object model for API endpoints

[UNIQUE] [Section 2 High] **Q58. Explain TestNG groups for API test organization. How do you run tests in parallel?**

- Explain TestNG groups for API test organization
- Implement parallel test execution

[UNIQUE] [Section 2 High] **Q60. What are best practices for API automation? How do you keep API tests maintainable?**

- What are API testing best practices?
- How do you ensure API tests are maintainable?
- What are some best practices for writing API tests with REST Assured?
- How do you ensure your API tests are maintainable?
- Best practices for API automation?

[UNIQUE] [Section 3 Medium] **Q100. How do you generate reports (Extent) for API tests?**

- Implement extent reports for API tests
- How to generate reports?

Postman Scripting & Newman (25 removed)

[UNIQUE] [Section 1 Critical] Q27. What are Collections, Environments and Variables in Postman? How do you use environment variables in scripts and import/export them?

- What are Postman Collections and why are they important?
- What are environments in Postman?
- What are variables in Postman and how do you use them?
- How do you import/export collections and environments in Postman?
- How do you use environment variables in scripts?

[UNIQUE] [Section 1 Critical] Q29. How do you write a basic test script in Postman (status code, response time, JSON body assertions)? Which scripting language does it use?

- How do you write a basic test script in Postman?
- What scripting language does Postman use for automation?
- How do you write assertions for status codes and response times?
- How do you validate JSON response body in Postman?

[UNIQUE] [Section 1 Critical] Q30. How do you chain requests in Postman (dynamic tokens, store and reuse values, capture header values, call one request from another, sequencing)?

- How do you handle dynamic data in Postman tests?
- How do you chain API requests in Postman?
- How do you handle dynamic tokens or session IDs in Postman?
- How do you store and use dynamic values across requests?
- How do you call one request from another in Postman?
- How do you capture values from headers in Postman?
- How do you handle API request sequencing in Postman?

[UNIQUE] [Section 2 High] Q61. What is Newman? How do you run Postman tests headlessly and generate HTML reports?

- What is Newman and why is it used? **[x2 exact repeat]**
- How do you run Postman tests headlessly?
- How do you generate HTML reports from Newman?

[UNIQUE] [Section 2 High] Q64. How do you run data-driven tests and parameterize requests in Postman?

- How do you run data-driven tests in Postman?
- How do you parameterize requests in Postman?

[UNIQUE] [Section 2 High] Q65. How do you validate JSON schema and arrays in Postman?

- How do you validate JSON schema in Postman?
- How do you validate a JSON schema in Postman tests?
- How do you validate arrays in Postman response?

[UNIQUE] [Section 2 High] Q66. How do you run only selected tests, folders or requests from a collection?

- How do you run only selected tests in a Postman collection?
- How do you run specific folders or requests from a collection in CI/CD?

[UNIQUE] [Section 2 High] Q67. How do you handle loops, conditional logic, random data and console logging in Postman scripts?

- How do you log output to Postman Console?
- How do you handle loops in Postman scripts?
- How do you generate random test data in Postman?
- How do you handle conditional testing in Postman scripts?

[UNIQUE] [Section 3 Medium] Q102. What is a Monitor in Postman and how do you schedule collections?

- What is a Monitor in Postman?
- How do you schedule Postman collections to run automatically?

[UNIQUE] [Section 3 Medium] Q104. How do you manage large collections and keep environments in sync across teams?

- How do you manage large Postman collections efficiently?
- How do you keep Postman environments in sync across teams?

CI/CD & Reporting for API Tests (5 removed)

[UNIQUE] [Section 1 Critical] Q31. How do you run API tests in CI/CD pipelines (Newman or REST Assured with Jenkins/Maven)? How do you trigger them after deployment?

- How do you automate Postman tests in CI/CD pipelines?
- How do you integrate Postman tests into CI/CD pipelines?
- How do you test APIs automatically after deployment?
- How do you integrate REST Assured tests with CI/CD pipelines?
- How to integrate REST Assured with Maven and Jenkins?
- How to trigger API tests in CI/CD?

Scenario-Based API Testing (11 removed)

[UNIQUE] [Section 1 Critical] Q32. How do you approach testing a new API when no documentation is available, or the requirements are incomplete and the behaviour changes often?

- How do you approach testing a new API when no documentation is available?
- A major new API endpoint has been delivered by development, but the requirements are incomplete and the endpoint behaviour changes every few days. How would you approach validation of this endpoint using automation (and manual) methods?

[UNIQUE] [Section 2 High] Q73. How do you test API pagination?

- How do you test API pagination? **[x2 exact repeat]**
- How to test pagination in APIs?
- How do you handle pagination in API testing using REST Assured?

[UNIQUE] [Section 2 High] Q75. How do you test APIs that return asynchronous responses?

- How do you test APIs that return asynchronous responses?
- How do you test APIs that return asynchronous results?

[UNIQUE] [Section 2 High] Q76. How do you test APIs that depend on other services, and in a microservices architecture?

- How do you test APIs that have dependencies on other services?
- How do you test APIs in a microservices architecture?

[UNIQUE] [Section 2 High] Q78. How do you validate API caching behaviour?

- How do you validate API caching behavior?
- How do you test if API caching is implemented correctly?

[UNIQUE] [Section 2 High] Q79. How do you test APIs under failure scenarios (including simulated network failures)?

- How do you test APIs under failure scenarios?
- How do you simulate network failures in API tests?

[UNIQUE] [Section 2 High] Q82. How do you handle API versioning in tests and ensure backward compatibility?

- How do you ensure API backward compatibility?
- How do you test versioned APIs?
- How do you handle versioning in API tests?
- How do you handle API versioning in REST Assured tests?

Reliability, Retry & Performance (6 removed)

[UNIQUE] [Section 1 Critical] Q33. How do you handle flaky or intermittent API tests? Implement retry logic for failed requests.

- How do you handle flaky APIs in automation?
- Implement retry logic for flaky API tests
- What strategies can you use to retry failed requests in REST Assured?
- How do you handle intermittent failures in API tests?

[UNIQUE] [Section 2 High] Q85. How do you handle timeouts, retries and error scenarios, and set a custom request timeout?

- How to handle timeouts, retries, and error scenarios in API testing?
- How do you test API timeout handling?
- How do you set a custom request timeout in REST Assured?

[UNIQUE] [Section 3 Medium] Q115. How do you perform load testing using REST Assured? How do you verify throughput and latency?

- How do you perform load testing using REST Assured?
- How do you verify performance metrics like throughput and latency in API tests?

Mocking & WireMock (7 removed)

[UNIQUE] [Section 2 High] Q86. What is the purpose and what are the benefits of mocking in API testing?

- What is the purpose of mocking in API testing?
- What are the benefits of using mocking in API testing?

[UNIQUE] [Section 2 High] Q87. How do you mock an API with WireMock (set up and start the server, configure a response, call it with REST Assured, stop it)?

- Test API with mock server (WireMock)
- How do you mock an API endpoint using REST Assured?
- How do you integrate REST Assured with WireMock?
- How do you set up and start a WireMock server for testing with REST Assured?
- How do you configure a mock response in WireMock?
- How do you use REST Assured to send requests to the WireMock server?
- How do you stop the WireMock server after tests are completed?

API Contract Testing (3 removed)

[UNIQUE] [Section 2 High] Q88. How do you validate API contracts? What is API contract testing and why is it important?

- How do you validate API contracts?
- What is API contract testing?
- Why is API Contract Testing important?

[UNIQUE] [Section 2 High] Q89. What tools are used for contract testing? What is Pact and how does it work?

- What are some tools commonly used for API Contract Testing?
- What is Pact, and how does it work in API Contract Testing?

CI/CD Interview Questions (61 removed)

Source file: CI/CD Interview Questions. Original 123 questions, 62 unique kept, 61 removed. Prioritized PDF: CICD_Interview_Questions_Prioritized.pdf.

CI/CD Concepts & QA's Role (16 removed)

[UNIQUE] [Section 1 Critical] Q1. What is CI/CD and why is it important in automation and for QA engineers?

- What is CI/CD, and why is it important in automation?
- What is CI/CD and why is it important for QA engineers?
- What is continuous integration (CI/CD) and how do you integrate tests?

[UNIQUE] [Section 1 Critical] Q2. What CI/CD tools have you used? What tools can be used for CI/CD in QA automation?

- What CI/CD tool you used
- What tools can be used for CI/CD in QA automation?

[UNIQUE] [Section 1 Critical] Q3. Explain the CI/CD pipeline and its common stages from a QA perspective.

- Explain CI/CD pipeline.
- What are the common stages of a CI/CD pipeline from a QA perspective?

[UNIQUE] [Section 1 Critical] Q4. What is the role of QA in a CI/CD pipeline, and how does CI/CD improve software quality?

- What is the role of QA in a CI/CD pipeline?
- How does CI/CD improve software quality for QA teams?

[UNIQUE] [Section 1 Critical] Q5. How do you integrate automated tests, scripts and your framework into a CI/CD pipeline? Is your framework CI/CD ready?

- Explain CI/CD integration with test automation frameworks.
- How do you integrate automated tests in a CI/CD pipeline?
- How do you integrate your framework with CI/CD pipeline?
- How do you integrate automation scripts into CI/CD?
- Framework CI/CD readiness

[UNIQUE] [Section 1 Critical] Q6. How do you integrate your Selenium suite with Jenkins, GitHub Actions or Azure DevOps?

- How do you integrate your Selenium suite with Jenkins or GitHub Actions?
- How do you integrate Selenium tests with Jenkins?
- How to integrate Selenium with CI/CD (both Jenkins and ADO)?

[UNIQUE] [Section 2 High] Q19. What is Continuous Integration (CI) and what is Continuous Delivery (CD)?

- What is continuous integration (CI)?
- What is continuous delivery (CD)?

[UNIQUE] [Section 2 High] Q21. How do you trigger a CI/CD pipeline automatically after code changes (Git triggers)?

- How do you trigger a CI/CD pipeline automatically after code changes?
- How do you configure Jenkins to trigger builds automatically from GitHub?
- Trigger builds using Git

[UNIQUE] [Section 3 Medium] Q45. How do you test a CI/CD pipeline, its rollback mechanism and its scheduling?

- How do you test CI/CD pipeline for automation tests?
- How do you test CI/CD rollback mechanism?
- How do you test Jenkins pipeline scheduling?

Jenkins: Jobs, Pipelines, Triggers & Scheduling (16 removed)

[UNIQUE] [Section 1 Critical] Q7. What is Jenkins and why is it important for QA? How and when do you use it in your automation?

- What is Jenkins and why is it important for QA engineers?
- What is Jenkins? **[x2 exact repeat]**
- How and when is Jenkins is used in your Automation?

[UNIQUE] [Section 1 Critical] Q8. What is the difference between a Freestyle job and a Pipeline job? Explain the basics of a Jenkins pipeline.

- Basics of Jenkins pipeline
- What's the difference between a Jenkins freestyle job and a pipeline script?
- What is the difference between a Freestyle job and a Pipeline job in Jenkins?
- Pipeline jobs vs freestyle jobs

[UNIQUE] [Section 1 Critical] Q9. What is a Jenkinsfile and why does QA use it?

- What is a Jenkinsfile and why QA uses it?
- What is the role of a Jenkinsfile?
- Jenkinsfile basics

[UNIQUE] [Section 1 Critical] Q10. How do you create and configure a Jenkins job to run (trigger) your automation suite?

- Triggering automation suite from Jenkins
- Jenkins - how to run jobs?
- How do you create a basic Jenkins job to run automation tests?
- How will you configure Jenkins job?
- How to set up a Jenkins job?

[UNIQUE] [Section 1 Critical] Q11. How do you parameterize a Jenkins build / pass parameters to a job?

- Passing parameters to Jenkins job
- How do you parameterize a Jenkins build for QA testing?

[UNIQUE] [Section 1 Critical] Q12. How do you schedule a nightly regression run in Jenkins (cron)?

- How do you schedule a nightly regression run in Jenkins?
- Scheduling builds (cron)

[UNIQUE] [Section 2 High] Q24. How do you run tests in parallel in Jenkins (parallel jobs)?

- How do you run parallel tests in CI/CD?
- How do you run tests in parallel in Jenkins?
- Running parallel jobs

Reports, Failures & Troubleshooting in CI (10 removed)

[UNIQUE] [Section 1 Critical] Q13. How are automation reports published and stored in Jenkins? How do you publish HTML and TestNG reports and share them?

- How are automation reports published in Jenkins?
- Where automation reports are stored or published
- How do you generate and share test reports in CI/CD?
- How do you publish HTML test reports in Jenkins?
- Generate TestNG reports in Jenkins

[UNIQUE] [Section 1 Critical] Q14. How do you handle failed builds and test failures in Jenkins?

- How do you handle failed builds in Jenkins for QA tests?
- How do you handle a failed automation build in Jenkins?
- Handling test failures

[UNIQUE] [Section 1 Critical] Q15. Tests pass locally but fail on Jenkins/CI nodes. How do you debug?

- Tests pass locally but fail on Jenkins (CI) nodes.
- Your tests pass locally but fail in CI/CD. How do you debug?

[UNIQUE] [Section 1 Critical] Q16. How do you handle flaky or unstable tests in CI/CD pipelines?

- How do you handle flaky tests in CI/CD pipelines?
- How do you deal with unstable test scripts in CI?

[UNIQUE] [Section 2 High] Q29. How do you archive test artifacts in Jenkins?

- How do you archive test artifacts in Jenkins?
- How to archive artifacts?

[UNIQUE] [Section 2 High] Q30. How do you set up email and Slack notifications after builds?

- Email notifications after builds
- Slack/email integration

Maven & Build Management (5 removed)

[UNIQUE] [Section 1 Critical] Q17. What is Maven and why do you use it in automation projects? What is pom.xml and which basic Maven commands do you use?

- What is Maven and why do you use it in automation projects?
- What is the use of pom.xml? (maven)
- Can you give some basic commands used in maven project?

[UNIQUE] [Section 1 Critical] Q18. What is the Maven build lifecycle?

- What is the Maven Build Lifecycle?
- Lifecycle of Maven

[UNIQUE] [Section 2 High] Q32. How do you manage different environments (QA, pre-production, production) with different data using Maven / pom.xml?

- How do you manage different environments using Maven?
- Consider you have to write test/suite for different environments(qa, preproduction, production) and pass different set of data for each environment. How will you do it using maven file(Pom.xml)?

[UNIQUE] [Section 2 High] Q33. How are dependencies handled in Maven at run time? If someone deleted all dependencies from the pom file, can you still execute tests?

- How will you handle dependencies in Maven at run time?
- Today we have executed some tests using maven, but tomorrow when you see that someone deleted all dependencies from .pom file then in that case will you be able to execute tests or not.

Docker for QA Automation (6 removed)

[UNIQUE] [Section 2 High] Q34. What is Docker and why is it useful in QA testing? How is it different from virtual machines?

- What is Docker and why is it useful in QA testing?
- Difference between Docker and Virtual Machines for QA?

[UNIQUE] [Section 2 High] Q36. How do you write a Dockerfile and run a Docker container for automation testing (including API testing)?

- How do you create and run a Docker container for automation testing?
- How to write a Dockerfile for QA automation projects?
- How do you use Docker for API testing?

[UNIQUE] [Section 2 High] Q37. How do you run Selenium in Docker and use Docker for cross-browser testing?

- How to run Selenium in Docker?
- How can Docker help in cross-browser testing for QA?

[UNIQUE] [Section 3 Medium] **Q50. What is a Docker Volume and how do you persist test results in containers?**

- How do you persist test results in Docker containers?
- What is a Docker Volume and why is it important for QA?

[UNIQUE] [Section 4 Low] **Q60. What is the difference between docker ps and docker images? How do you remove unused containers and images?**

- What is the difference between docker ps and docker images?
- How do you remove unused containers and images to save disk space?

Linux / CLI Basics for QA (8 removed)

[UNIQUE] [Section 3 Medium] **Q57. How do you see running processes and check disk and memory usage?**

- How do you see running processes?
- How do you check disk space usage?
- How do you check memory usage?

[UNIQUE] [Section 3 Medium] **Q58. How do you find a file in Linux and check network connectivity?**

- How do you find a file in Linux?
- How do you check network connectivity?

[UNIQUE] [Section 4 Low] **Q61. How do you list files, check and change the current directory, and create directories and files?**

- How do you list files and directories in Linux?
- How do you check your current directory?
- How do you change directories?
- How do you create a new directory?
- How do you create a new file from CLI?

[UNIQUE] [Section 4 Low] **Q62. How do you view file contents, and copy, move and delete files?**

- How do you view the contents of a file?
- How do you copy, move, and delete files in Linux?

Git and Maven Questions (27 removed)

Source file: Git and Maven Questions. Original 68 questions, 41 unique kept, 27 removed. Prioritized PDF: Git_and_Maven_Questions_Prioritized.pdf.

Git: Merge, Rebase, Conflicts & Pull Requests (5 removed)

[UNIQUE] [Section 1 Critical] Q1. What is the difference between git merge and git rebase? When do you use each?

- Git: merge vs rebase? Handling merge conflicts?
- Difference between git merge and git rebase.

[UNIQUE] [Section 1 Critical] Q2. How do you resolve merge conflicts in Git?

- How do you resolve merge conflicts in Git?
- Resolving merge conflicts

[UNIQUE] [Section 1 Critical] Q3. What is a pull request (PR) and why does QA use it? How do you create one?

- What is a pull request (PR) and why QA uses it?
- Creating pull requests

[UNIQUE] [Section 1 Critical] Q4. What is the difference between git fetch and git pull?

- What is the difference between git fetch and git pull?
- Difference between fetch and pull

[UNIQUE] [Section 2 High] Q18. How do you stash changes in Git and when do you use git stash?

- How to stash changes in Git?
- git stash usage

Git: Fundamentals, Branching & Workflow (5 removed)

[UNIQUE] [Section 1 Critical] Q5. What is Git and why is it important for QA engineers? What is the purpose of a version control tool?

- What is Git and why is it important for QA engineers?
- What is the purpose of version control tool?
- What is Git and why use it?

[UNIQUE] [Section 1 Critical] Q8. Which Git commands have you used?

- Git commands you have used
- What are the git commands you have used?

[UNIQUE] [Section 2 High] Q21. What is .gitignore and why does QA use it?

- What is .gitignore and why QA uses it?
- .gitignore file

[UNIQUE] [Section 3 Medium] Q37. How do you fork a repository and set an upstream remote?

- Forking a repository
- Setting upstream remote

Git: Everyday Commands, Undo & History (7 removed)

[UNIQUE] [Section 1 Critical] Q9. How do you revert a commit without losing history?

- How do you revert a commit without losing history?
- Reverting a commit

[UNIQUE] [Section 2 High] **Q23. What are the basic Git commands (init, clone, status, add, commit, push, pull)? How do you clone a repository from GitHub?**

- How to clone a repository from GitHub?
- git init, git clone, git status
- git add, git commit
- git push, git pull

[UNIQUE] [Section 2 High] **Q24. How do you create and switch to a new branch, and check which branch you are on?**

- How do you create and switch to a new branch in Git?
- Creating and switching branches
- How do you check which branch you are on?

[UNIQUE] [Section 3 Medium] **Q38. How do you delete or rename a branch?**

- How do you delete a branch in Git?
- How to rename a branch in Git?

Maven: Fundamentals, pom.xml & Dependencies (6 removed)

[UNIQUE] [Section 1 Critical] **Q12. What is Maven and why is it used in automation frameworks?**

- What is Maven?
- What is Maven and why is it used in automation frameworks?

[UNIQUE] [Section 1 Critical] **Q13. What is pom.xml and what is its structure?**

- Structure of pom.xml
- What is pom.xml in Maven?

[UNIQUE] [Section 1 Critical] **Q14. How does Maven manage dependencies? What are dependency scopes and how do you add a dependency from Maven Central?**

- How does Maven manage dependencies?
- Scope of dependencies
- Adding dependencies from Maven Central

[UNIQUE] [Section 2 High] **Q29. What are Maven repositories and their types (local, central, remote)?**

- What is a repository in Maven?
- What are Maven repositories and their types?

[UNIQUE] [Section 2 High] **Q30. What is a parent POM and how does inheritance work in Maven?**

- What is a parent POM?
- Inheritance in Maven

Maven: Lifecycle, Commands & Plugins (4 removed)

[UNIQUE] [Section 1 Critical] **Q15. What is the Maven lifecycle and what are its phases?**

- Maven lifecycle?
- Life cycle phases in Maven
- What is a Maven lifecycle?

[UNIQUE] [Section 2 High] **Q32. What is the difference between build and clean, and between install and deploy?**

- What is the difference between build and clean in Maven?
- What is the difference between install and deploy phase in Maven?

[UNIQUE] [Section 2 High] **Q33. What is the purpose of Maven plugins? What is the Surefire plugin?**

- What is the purpose of Maven plugins?
- Maven Surefire Plugin

Non-Technical Interview Questions (3 removed)

Source file: Non-Technical Interview Questions. Original 27 questions, 24 unique kept, 3 removed. Prioritized PDF: Non_Technical_Interview_Questions_Prioritized.pdf.

Introduction, Role & Project Questions (1 removed)

[UNIQUE] [Section 1 Critical] Q2. What are your roles and responsibilities in your current assignment?

- What are your Roles and Responsibilities?
- What are your Roles & Responsibilities in your current Assignment?

Career Move, Motivation & Self-Assessment (1 removed)

[UNIQUE] [Section 2 High] Q14. How quickly can you learn new technologies and processes? How interested are you in learning new technologies?

- How quickly can you learn new technologies and processes?
- How interested are you in learning new technologies?

Automation Strategy & ROI (1 removed)

[UNIQUE] [Section 2 High] Q22. How do you demonstrate automation ROI to leadership? What is your strategy to rapidly improve ROI?

- Strategy to rapidly improve ROI in automation?
- Demonstrate automation ROI to leadership.